

Penalized Estimation of Moderator Effect Networks for High-Dimensional Multiple Subject Longitudinal Data

José Á. Sánchez Gómez Holly O’Rourke Gene Brewer *

Abstract

In psychological research, there is a growing interest in the quantification of both nomothetic and idiographic variation in human behavior. This is especially pressing in the analysis of intensive longitudinal data for multiple subjects, where there is interest in understanding both inter-subject and intra-subject variability for a set of d variables evolving over time. Often, researchers measure a set of p additional time-invariant moderators, which further describe the subjects in the study. Although several methods have been recently proposed for the simultaneous estimation of common and individual components of variation for longitudinal data, they are not designed to incorporate the information of the time-invariant moderators, which may have a role in the intra-subject variability. Current methods that may incorporate moderator variables are not properly equipped to handle high-dimensionality when the number of variables of interest d and moderator variables p are large. In this paper, we propose the MOD-VAR method, that efficiently estimates the presence of common variation across subjects, as well as individual variability explained by underlying moderator variables of interest. We provide numerical simulations that confirm the superior performance of our proposed MOD-VAR with methods in the literature in terms of network recovery and forecasting accuracy. Our theoretical guarantees show that our proposed MOD-VAR method estimates both the joint and individual network connections consistently, even for limited number of subjects n and measured time points T , as long as $nT \gtrsim \log(pd)$. Finally, we exemplify the use of our proposed MOD-VAR on a longitudinal study of human behavior, and study the effect that moderators like gender or employment have on the dynamics of anxiety, depression, and sleep.

Keywords: [add](#)

*José Á. Sánchez Gómez is Assistant Professor, Department of Statistics, University of California Riverside, Riverside, CA 92521, USA. E-mail: josesa@ucr.edu. Holly O’Rourke is Associate Professor, Department of Psychology, University of California Riverside, Riverside, CA 92521, USA. Email: holly.orourke@ucr.edu. Gene Brewer is Professor, Department of Psychology, University of California Riverside, Riverside, CA 92521, USA.

Corresponding Authors: José Á. Sánchez Gómez (josesa@ucr.edu), Holly O’Rourke (holly.orourke@ucr.edu).

1 Introduction

Since the manifesto of Peter Molenaar [cite](#), psychological researchers have become increasingly aware of the importance of intra-subject variability when studying human behavior. Before the aforementioned manifesto, many psychological studies focused on the study of inter-subject variation, also known as nomothetic components of variation in human behavior. There is now growing recognition that, in many scenarios, conclusions reached on groups may not be applicable to the individual, due to the variability that behavior may have at the individual level, also known as idiographic variation. It is vital that modern methods in psychological research successfully acknowledge the presence of both sources of variation when studying human behavior.

The need for methods that can estimate both nomothetic and idiographic components are especially important when analyzing intensive longitudinal data (ILD). Thanks to the advent of mobile health monitoring devices, there is a renewed interest in the collection of intensive time series data measuring a wide range of behavioral variables. This data contains relevant insight with both heterogeneity at the individual level, as well as cross-subject variability. Designing models that are capable of distinguishing the common patterns across individuals, as well as the within-subject variability is of vital interest to psychological research.

In this context, one important exploration is to determine how current psychological and behavioral variables are influenced by past values. One popular approach to model this is through the use of vector autoregressive models (VAR), in which each variable of interest is represented with a node, and a directed link between two nodes represents a lagged effect of one variable to another. The VAR model has been deployed for the analysis of [this](#), [this](#), [this](#). Specifically, in psychology it has been used for the analysis of [this](#) [this](#) [this](#).

When encountering multiple subject data, there are several approaches that may be taken to account for the nomothetic and idiographic variability in the data. One simple approach is pool the data across subjects, to obtain an overall estimate of the network of lagged effects aimed to be valid across all subjects [cite](#). This approach clearly focuses on the nomothetic component of variation, and completely ignores the potential for network effects that are individual to each subject. Alternatively, one may perform the estimation of all

networks separately, by fitting a VAR model for each subject separately. This approach aims to recover the idiographic components. While desirable, this approach ignores the potential common structure across subjects, leading to inaccurate estimation of the network effects.

Recently, several methods have been proposed for the recovery of both nomothetic and idiographic components for intensive longitudinal data. the GIMME method and its extensions [4] use a sequential testing approach, combined with a consensus rule to select the common network structure across multiple subjects. On the other hand, the Multi-VAR method [3] introduces a penalized approach that achieves the estimation of the common network, as well as subject specific network connections via the optimization of an ℓ_1 -penalized loss minimization problem. We emphasize that, for both the GIMME and Multi-VAR, the idiographic structure is allowed to be arbitrary, and does not aim to analyze the origin of this variability, or whether it is connected to underlying moderators. Thus, methods that do not only measure but aim at *explain* the origins of the idiographic variability of subjects are vital for future psychological research.

When gathering intensive longitudinal data, researchers often gather additional time-invariant moderator variables that further describe the subjects in the study. In the case of observational studies, researchers may consider factors like age, socio-economic status, ethnicity or clinical history. In experimental settings, there may be interest in distinguishing the effect of a particular set of treatments on subject behavioral dynamics. Thus, a natural question of interest to behavioral researchers may be: *“are the individual related or explained by these underlying moderators of interest?”* This question is at the core of the present paper.

Models for relating VAR coefficients to additional time-invariant variables have been explored in the past, and belong to the area of multi-level VAR [cite](#). These methods suffer from several main limitations: these often assume that the, and (i) these methods are often unable to handle a large number of variables d or moderator variables p , restricting, unless the number of subjects or time-measurements per subjects are prohibitively large. Thus, there is a pressing need for methods that can estimate the influence that moderator variables have on the intra-subject variability of behavior, and can provide reliable estimates, even in cases where the number of parameters may be significantly larger than the number of observations.

In our present paper, we propose the moderator vector autoregressive model (MOD-VAR), which simultaneously quantifies the presence of common nomothetic structure, as well as idiographic structure linked to moderator variables for in multiple subject longitudinal data. We achieve this through a penalized optimization approach based on the famous least absolute shrinkage and selection operator (LASSO) [cite](#). Our method is designed to accurately recover the presence of inter-subject heterogeneity linked to moderator variables, even for small sample sizes n with large number of moderators n and time measurements T .

Our theoretical findings confirm that our proposed MOD-VAR can achieve consistent estimation of all parameters, as long as $nT \gg \log(pd^2)$. Furthermore, we conduct extensive simulation scenarios, which show the improved performance of our proposed MOD-VAR when compared to other methods in the literature designed for the estimation of nomothetic and idiographic components for multiple subject longitudinal data. To showcase the usefulness of our method in real data applications, we analyze [the daily emotional assessment dataset](#), and show links between gender, employment status and the dynamics of depression, anxiety and sleep.

The rest of the paper is organized as follows. In section 2 we introduce our model of moderator time series model, and introduce our MOD-VAR estimator. In section 3 we provide details on the implementation of our proposed method. In Section 5 we provide numerical comparisons of our proposed MOD-VAR with methods found in the literature. In Section 6 we exemplify the use of our MOD-VAR method in the analysis of daily mood and behavior psychological data. Closing remarks and discussions are found in Section 7

2 Moderator Vector Autoregressive Model

2.1 Vector Autoregressive Models

Consider a longitudinal study with n subjects. For each subject k , we have access to the measurements of d different variables of interest over T_k equally spaced time points, which we denote by $\mathbf{Y}_k = \{\mathbf{Y}_t^{(k)} : 1 \leq t \leq T_k\}$. In this context, it is important to understand the lagged temporal relationships that exist between the different variables. This can be

achieved via a VAR(1) model, which can be expressed as

$$\mathbf{Y}_t^{(k)} = \Phi^{*(k)} \mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 2 \leq t \leq T_k,$$

where $\Phi^{*(k)}$ is a $d \times d$ matrix that represents the true lag-1 multivariate relationships for the subject k . In particular, for some $1 \leq i, j \leq d$, the entry $\Phi_{ij}^{*(k)}$ represents the effect that the variable j at time $(t-1)$ affects the variable i at time t . Notice that, for each subject k , we define a distinct coefficient matrix $\Phi^{*(k)}$.

When encountering a study with multiple subjects, there are several potential approaches to consider. A simple approach consists of simply assuming that all subjects have the same lagged temporal relationships, *i.e.*, $\Phi^* := \Phi^{*(1)} = \Phi^{*(2)} = \dots = \Phi^{*(n)}$. Under this assumption, the estimation of Φ can be performed by pooling the data from all subjects and performing standard VAR estimation. This approach exclusively focus on the estimation of nomothetic components, missing the opportunity to understand relevant idiographic descriptions of subjects' behavior. Alternatively, one may perform the estimation of each $\{\Phi^{*(1)}, \Phi^{*(2)}, \dots, \Phi^{*(n)}\}$ separately, by fitting a different VAR model to each model. While this approach allows for the study of idiographic structure, it does not exploit the potential common structure that is likely to be shared across most subjects.

More recently, Multi-VAR Fisher et al. [3] explores the estimation of multiple VAR models, allowing for the presence of both nomothetic and idiographic components. To this end, they assume that the subject-level coefficients of the VAR(1) model satisfy the decomposition,

$$\Phi^{*(k)} = \Phi^{*(0)} + \Delta^{*(k)}, \quad \text{for all } k = 1, \dots, n.$$

where $\Phi^{*(0)}$ is a $d \times d$ matrix representing the coefficients common across all subjects, and $\Delta^{*(k)}$ represents relationships exclusive to subject k . Then, we can interpret $\Phi^{*(0)}$ and $\Delta^{*(k)}$ as the nomothetic and idiographic components, respectively.

2.2 Moderator Vector Autoregressive Models

Despite the flexibility of methods in the literature for quantifying both nomothetic and idiographic components [4; 3], there is still an important question that remains unanswered: *Can part of the behavioral heterogeneity found across subjects be partially or fully explained by external moderator variables?* This question is at the core of our proposed methodology.

Here, we consider that, for each subject $k = 1, 2, \dots, n$, in addition to the time-series data $\mathbf{Y}_k$, we have a set of p potential moderator variables $\mathbf{X}_k = (X_{k1}, X_{k2}, \dots, X_{kp})'$ arranged in a $p \times 1$ column vector. Notice that the moderators in $\mathbf{X}_k$ are not time-varying, and thus can be thought as stable features of each individual. These moderators can be clinical or demographic variables such as gender or age. We may also consider control experimental variables of interest such as treatment status of individual, or magnitude of the dose provided to a patient.

First, consider the scenario in which the cross-subject heterogeneity can be fully explained by the set of external moderators $\mathbf{X}_k$. We can represent the moderator-dependent time series lagged relationships as $\Phi^{*(k)} = \Phi^*(\mathbf{X}_k)$, where, for any vector of p moderators $\mathbf{x} = (x_1, x_2, \dots, x_p)'$,

$$\Phi^*(\mathbf{x}) := \Phi^{*(0)} + x_1\Phi^{*(M1)} + x_2\Phi^{*(M2)} + \dots + x_p\Phi^{*(Mp)}, \quad (1)$$

where $\Phi^{*(0)}, \Phi^{*(M1)}, \Phi^{*(M2)}, \dots, \Phi^{*(Mp)}$ are all $d \times d$ matrices. Similar to the proposal of the multi-VAR, Φ^0 represents the shared relationship dynamics. Furthermore, for each moderator variable $1 \leq \ell \leq p$, the matrix $\Phi^{*(M\ell)}$ represents the effect that the ℓ -th moderator has on the time series dynamic. More specifically, for $1 \leq i, j \leq d$ and $1 \leq \ell \leq p$ the value of $\Phi_{ij}^{*(M\ell)}$ represents the change that the moderator variable ℓ has on the lagged relationship of j on the variable i . Notice that, if a subject has $x_\ell = 0$, then $x_\ell\Phi^{*(M\ell)} = \mathbf{0}$ and thus, the moderator ℓ does not have any effect on the multivariate time dynamics for this subject. In addition, the larger the value of x_ℓ , the stronger the effect that the moderator ℓ have on the dynamic of the corresponding subject.

Under the model (1), the time series dependence for subject k can be written as,

$$\mathbf{Y}_t^{(k)} = \Phi(\mathbf{X}_k)\mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\epsilon}_t^{(k)}, \quad \text{for all } 1 \leq k \leq n \text{ and } 2 \leq t \leq T_k. \quad (2)$$

Example 1. We consider a VAR(1) model with n subjects. For each subject, we measure two time-invariant moderators (i.e., $p = 2$). In that case, $\mathbf{X}_k = (X_{k1}, X_{k2})'$. Then, the MOD-VAR model with 2 moderators is given by,

$$\begin{aligned} \mathbf{Y}_t^{(k)} &= \Phi(\mathbf{X}_k)\mathbf{Y}_{t-1}^{(k)} + \mathbf{e}_t^{(k)} \\ &= [\Phi^{*(0)} + X_{k1}\Phi^{*(M1)} + X_{k2}\Phi^{*(M2)}] \cdot \mathbf{Y}_{t-1}^{(k)} + \mathbf{e}_t^{(k)} \end{aligned}$$

Notice that each subject k will have their own VAR structure. This structure will incorporate a common structure shared across all subjects $\Phi^{*(0)} \in \mathbb{R}^{d \times d}$. It also has individual structure.

Now, instead of having this individual structure vary arbitrarily, it has the form $X_{k1}\Phi^{*(M1)} + X_{k2}\Phi^{*(M2)}$. Therefore, it depends on the information of our moderator variables $\mathbf{X}_k = (X_{k1}, X_{k2})'$ of each subject. A further description of this scenario can be found in Figure 3

2.3 Moderator Vector Autoregressive Models with Fully Idiographic Components

We may consider a further scenario, in which the moderator variables do not explain all cross-subject heterogeneity. In this case, we consider that, in addition to a shared network across all individuals Φ , and moderator-related individual structure, there is an additional fully idiographic structure in our model. In that case, we consider the following additional modeling approach, where for each subject k , the coefficient matrix $\Phi^{(k)}$ can be decomposed as

$$\Phi^{(k)} = \Phi(\mathbf{X}_k) + \Delta^{(Ik)}, \quad (3)$$

where $\Delta^{(Ik)}$ is a $d \times d$ matrix that contains the fully idiographic structure of each subject. Notice that the term $\Phi(\mathbf{X}_k)$ contains both the shared structure across all subjects, and the relationships that can be explained by external moderator variables. The remaining term $\Delta^{(Ik)}$ serves to model any remaining significant individual relationships. Then, we can rewrite the time series dependence as,

$$\mathbf{Y}_t^{(k)} = (\Phi(\mathbf{X}_k) + \Delta^{(Ik)})\mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 1 \leq k \leq n \text{ and } 2 \leq t \leq T_k, \quad (4)$$

Example 2. *An example where we include discrete grouping variables, like "gender", "treatment", and "age". Emphasize that we can incorporate continuous, discrete covariates. Modify figure to incorporate to this example.*

Our goal in this work is not only to estimate the nomothetic component Φ^0 , but also the matrices $\{\Phi^{(M\ell)}\}_{\ell=1}^p$ representing the effects that the underlying moderator variables have on the individual behavior dynamic, and $\{\Delta^{(Ik)}\}_{k=1}^n$ the fully idiographic terms that cannot be explained by the underlying moderators. In the following section, we introduce our MOD-VAR estimation method, and provide implementation details.

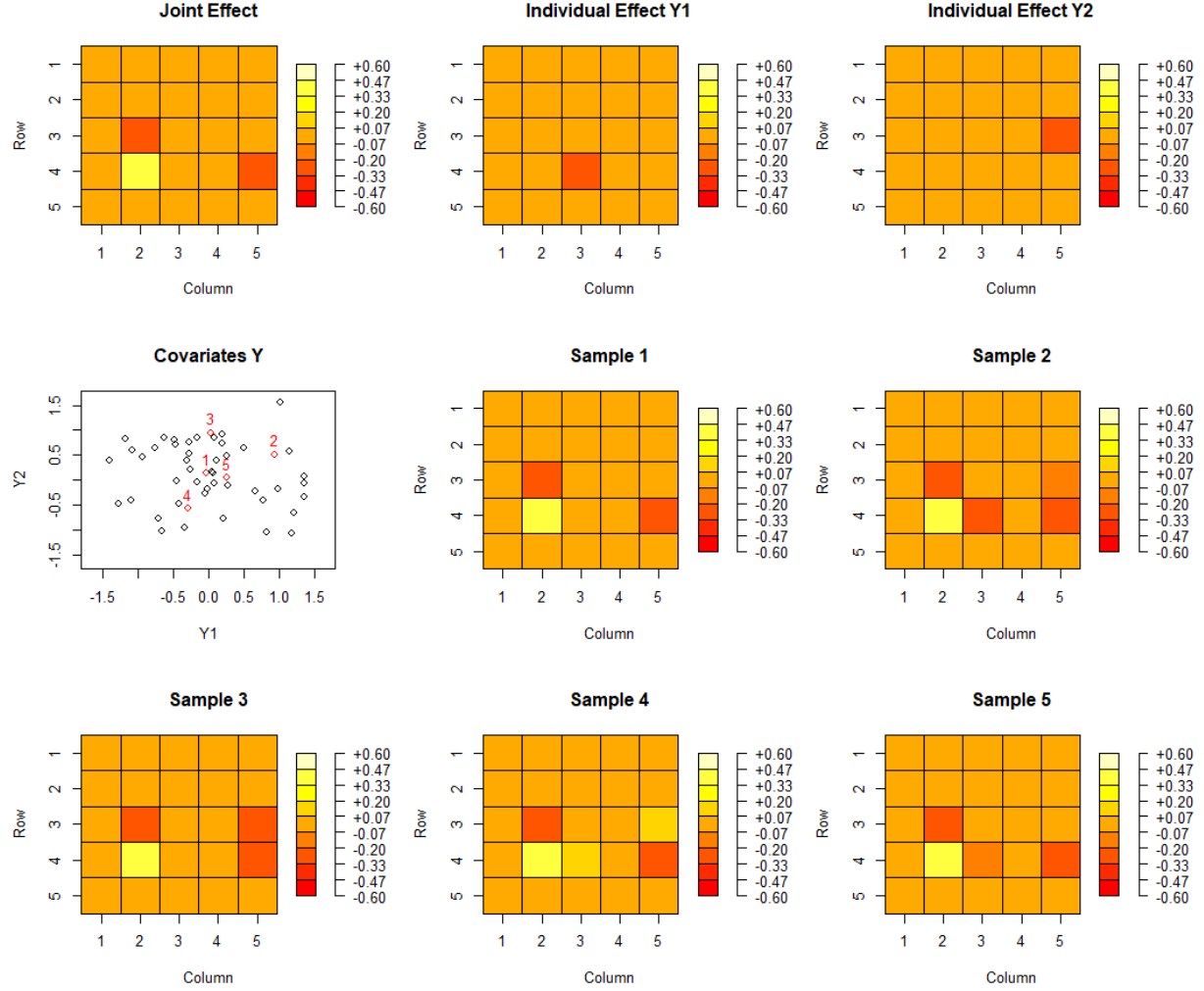


Figure 1: Illustration of the Regression VAR framework with 2 covariate-dependent effects, as discussed in Example 1. Top-panels: visualization of the joint effects that are common across all subjects. We also visualize the effects that are individual to variables X_1 and X_2 . Center-left panel: Visualization of the covariate information for $N = 50$ subjects. We highlight the covariate information of the first 5 subjects. Remaining panels: Visualization of the effects for the first 5 subjects. We observe that, as subject 1 has nearly zero for both X_1 and X_2 , the effects are mostly just the joint effects. Subject 3 has a high X_2 covariate, but low X_1 covariate, and therefore its effects are a mixture of the joint and X_2 effects. Subject 2 has large covariates for both X_1 and X_2 , so its time series effects are a mixture of all.

3 Methodology

We introduce our method for simultaneously estimating the shared lagged relationships across all subjects Φ^0 , as well as the moderated effects $\{\Phi^{(M\ell)}\}_{\ell=1}^p$ and fully idiographic relationships $\{\Delta^{(Ik)}\}_{k=1}^n$. We achieve this via an ℓ_1 -penalized time series estimation approach, which takes into account the estimation of all components. We first introduce the method when no fully idiographic component is present, *i.e.*, $\Delta^{(I1)} = \Delta^{(I1)} = \dots = \Delta^{(In)} = \mathbf{0}$.

3.1 Mod-VAR Only Moderator Components

Consider the case in which all inter-subject variability can be fully explained by our moderator variables, as described in (1). Then, our time series satisfies the relationship based on equation (4). Consider $\Phi = [\Phi^{(0)}, \Phi^{(M1)}, \Phi^{(M2)}, \dots, \Phi^{(Mp)}]$, and, for any vector of p moderators $\mathbf{x}$, we define $\Phi(\mathbf{x})$ as the $d \times d$ matrix given by

$$\Phi(\mathbf{x}) := \Phi^{(0)} + x_1\Phi^{(M1)} + x_2\Phi^{(M2)} + \dots + x_p\Phi^{(Mp)}. \quad (5)$$

We aim at providing an estimation $\hat{\Phi}$ for the true nomothetic lagged relationships $\Phi^{*(0)}$, as well as the moderator effects $\{\Phi^{(M\ell)}\}_{\ell=1}^p$. Notice this requires the estimation of a total of $q = pd^2$ parameters. This can be challenging, specially in cases where the number of study subjects n or number of time points per subject $\{T_k\}_{k=1}^n$ are small. In such scenarios, one may estimate Φ^* under the assumption of sparsity, *i.e.*, only a small number of parameters in Φ^* are non-zero, with the vast majority of the entries in Φ^* being zero. As seen in the literature, the estimation of such sparse VAR structures can be achieved through the use of ℓ_1 -penalties [1; 3]. Here, we adapt such sparse-estimation techniques for detecting moderator effects in the VAR models of interest.

To derive a sparse estimation of Φ^* , we introduce the MOD-VAR optimization problem

$$\hat{\Phi} = \underset{[\Phi^0, \Phi^{(M1)}, \dots, \Phi^{(Mp)}]}{\operatorname{argmin}} \frac{1}{2N} \sum_{k=1}^n \sum_{t=2}^{T_k} \|\mathbf{Y}_t^{(k)} - \Phi(\mathbf{X}_k)\mathbf{Y}_{t-1}^{(k)}\|_2^2 + \lambda_1 \|\Phi^0\|_1 + \lambda_2 \sum_{\ell=1}^p \|\Phi^{(M\ell)}\|_1. \quad (6)$$

where $N = T_1 + T_2 + \dots + T_n - n$. We can simplify the expression of (6) to a reduced

matrix form. For this, consider

$$\mathbf{Z}^k = \begin{pmatrix} (\mathbf{Y}_T^k)' \\ (\mathbf{Y}_{T-1}^k)' \\ \vdots \\ (\mathbf{Y}_2^k)' \end{pmatrix}, \quad \mathbf{Y}^k = \begin{pmatrix} (\mathbf{Y}_{T-1}^k)' \\ (\mathbf{Y}_{T-2}^k)' \\ \vdots \\ (\mathbf{Y}_1^k)' \end{pmatrix}, \quad \mathbf{E}^k = \begin{pmatrix} (\boldsymbol{\varepsilon}_T^k)' \\ (\boldsymbol{\varepsilon}_{T-1}^k)' \\ \vdots \\ (\boldsymbol{\varepsilon}_2^k)' \end{pmatrix}. \quad (7)$$

Here the matrices $\mathbf{Z}^k$, $\mathbf{Y}^k$ and $\mathbf{E}^k$ aggregate the contemporaneous, lagged and error terms for subject k respectively. To incorporate the dependency on moderators, we introduce

$$\mathbf{W}^k = [\mathbf{Y}^k, X_{k1}\mathbf{Y}^k, \dots, X_{kp}\mathbf{Y}^k] = \mathbf{X}'_k \otimes \mathbf{Y}^k \in \mathbb{R}^{(T_k-1) \times (pd)}.$$

Then, one can show that $(\mathbf{Y}^k)' \cdot \Phi^*(\mathbf{X}_k) = (\mathbf{W}^k) \cdot (\Phi^*)'$. Now, we gather the data from all subjects into a single object by defining

$$\mathbf{Z} = \begin{pmatrix} \mathbf{Z}^1 \\ \mathbf{Z}^2 \\ \vdots \\ \mathbf{Z}^n \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} \mathbf{W}^1 \\ \mathbf{W}^2 \\ \vdots \\ \mathbf{W}^n \end{pmatrix}, \quad \mathbf{E} = \begin{pmatrix} \mathbf{E}^1 \\ \mathbf{E}^2 \\ \vdots \\ \mathbf{E}^n \end{pmatrix}. \quad (8)$$

Then, we can resume the entire model (1) to the simplified matrix notation $\mathbf{Z} = \mathbf{W}(\Phi^*)' + \mathbf{E}$.

Furthermore, the optimization problem (6) is equivalent to the matrix version

$$\hat{\Phi} = \underset{\Phi=[\Phi^0, \Phi^{(M1)}, \dots, \Phi^{(Mp)}]}{\operatorname{argmin}} \left\| \mathbf{Z} - \mathbf{W}\Phi' \right\|_F^2 + \lambda_1 \left\| \Phi^0 \right\|_1 + \lambda_2 \left\| [\Phi^{(M1)}, \dots, \Phi^{(Mp)}] \right\|_1. \quad (9)$$

Then, if we define

$$\hat{\Gamma} = \mathbf{I}_d \otimes \mathbf{W} = \begin{pmatrix} \mathbf{W} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{W} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{W} \end{pmatrix}; \quad \boldsymbol{\beta} = \operatorname{vec}(\Phi') = \operatorname{vec} \begin{pmatrix} \Phi'_1 \\ \Phi'_2 \\ \vdots \\ \Phi'_d \end{pmatrix}; \quad y = \operatorname{vec}(\mathbf{Z});$$

Then, we can write our optimization (9) as a single LASSO optimization problem of the form,

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \frac{1}{2N} \|y - \hat{\Gamma}\boldsymbol{\beta}\|_2^2 + \sum_{i=1} \lambda_i |\beta_i|, \quad (10)$$

where $\lambda_i = \lambda_1$ if the entry β_i corresponds to a shared component in Φ^0 and $\lambda_i = \lambda_2$ if it corresponds to a moderator effect. Notice that (10) can be given directly to a solver of the LASSO optimization problem, such as the implementation found in the `glmnet` package in R. Our implementation leverages this fact to solve (10) through the `glmnet` package. Once $\hat{\boldsymbol{\beta}}$ is obtained, we can rearrange from vector to matrix form, to recover the matrix

$\hat{\Phi} = [\hat{\Phi}^0, \hat{\Phi}^{(M1)}, \dots, \hat{\Phi}^{(Mp)}]$. Observe that this model performs the sparse estimation of a total of $(p+1)d^2$ parameters.

Observe that the objects $y, \hat{\Gamma}$ can be constructed based on the time series data $\{\mathbf{Y}_k : k = 1, \dots, n\}$ and the matrix of moderator data,

$$\mathbf{X}_M = \begin{pmatrix} \mathbf{X}'_1 \\ \mathbf{X}'_2 \\ \vdots \\ \mathbf{X}'_p \end{pmatrix} = \begin{pmatrix} X_{11} & X_{12} & X_{13} & \dots & X_{1p} \\ X_{21} & X_{22} & X_{23} & \dots & X_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & X_{n3} & \dots & X_{np} \end{pmatrix}.$$

As we observe, variations on the design of the matrix of moderator data $\mathbf{X}$ can help estimate fully idiographic and moderator/idiographic mixed models.

3.2 Mod-VAR with Only Fully Idiographic Components

Consider the case in which we possess no moderator data, and we are exclusively interested in studying the presence of common and individual structure in the time series data. This scenario corresponds to the problem studied by the multi-VAR method [3]. It can be shown that the algorithm described in Section 3.1 can be directly used to perform this estimation. For each subject k , we define the artificial moderator vector $\mathbf{X}_k^e = (0, \dots, 0, 1, 0, \dots, 0)'$, a column vector of size $n \times 1$, with zero in all entries except the k -th entry. Then, observe that, when considering $\mathbf{X}_k^e$ as a moderator vector into (5)

$$\Phi(\mathbf{X}_k) = \Phi^0 + \Phi^{(Mk)}, \quad \text{for all } k = 1, 2, \dots, n.$$

Thus, the matrix $\Phi^{(Mk)}$ exclusively appears on the coefficients of subject k , and thus is equivalent to a fully idiographic component $\Delta^{(Ik)}$ for subject k . Under this consideration, the MOD-VAR only considering idiographic components can be solved via the vectorization described in Section 3.1. In other words, we can fit our MOD-VAR model considering only nomothetic and fully idiographic components via the moderator data matrix,

$$\mathbf{X}_I = \begin{pmatrix} (\mathbf{X}_1^e)' \\ (\mathbf{X}_2^e)' \\ \vdots \\ (\mathbf{X}_n^e)' \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}. \quad (11)$$

The optimization problem (9) solved for the moderator design matrix (11) is equivalent to the multi-VAR model proposed in [3], and requires the estimation of a total of $(n + 1)d^2$ parameters.

3.3 Mod-VAR with Moderator and Fully Idiographic Components

We may be interested in the most general modeling scenario, where the lagged relationships have all joint, moderator-based and fully idiographic components. The estimation of all these components can be obtained via a combination of the approaches used in Section 3.1 and 3.2.

Consider, for n subjects, the $n \times p$ moderator data matrix $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]'$. Now, consider the expanded matrix

$$\mathbf{X}_{I,M} = \begin{pmatrix} \mathbf{X}_M & \mathbf{X}_I \end{pmatrix} = \begin{pmatrix} X_{11} & X_{12} & X_{13} & \dots & X_{1p} & 1 & 0 & 0 & \dots & 0 \\ X_{21} & X_{22} & X_{23} & \dots & X_{2p} & 0 & 1 & 0 & \dots & 0 \\ X_{31} & X_{32} & X_{33} & \dots & X_{3p} & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & X_{n3} & \dots & X_{np} & 0 & 0 & 1 & \dots & 1 \end{pmatrix}.$$

Notice that, by incorporating both moderator-dependent and fully idiographic components to our moderator data matrix $\mathbf{X}$, we can estimate all the parameters $\Phi^{(0)}$, $\{\Phi^{(M\ell)}\}_{\ell=1}^p$ and $\{\Delta^{(Ik)}\}_{k=1}^n$ simultaneously by solving the problem (10) with the appropriate definition of $\hat{\Gamma}$ and y . Observe that in this scenario, we aim at providing a sparse estimate of a total of $(n + p + 1)d^2$ parameters.

3.4 Tuning Parameter Selection

One of the main concerns when fitting a penalized model is to determine the best procedure for selecting tuning parameters. In particular, due to the dependence present in time series data, direct sample splitting is not feasible, leading to sophisticated procedures of data splitting to obtain a reasonable cross-validation procedure. For example [add references, and what they do.](#)

3.4.1 Cross-Validation

Contrary to other methods like GIMME and Multi-VAR, where the individual one can generate “predictive model coefficients”: with a model $\hat{\Psi} \in \mathbb{R}^{d \times (p(d+1))}$ trained on the data of N individuals $\{(\mathbf{X}_k, \mathbf{Y}^{(k)})\}_{k=1}^N$, we may provide an estimate of the coefficients of an out-of-sample individual $(\mathbf{X}_{N+1}, \mathbf{Y}^{(N+1)})$ via $\hat{\Phi}_{N+1} = \hat{\Psi}_0 + \sum_{i=1}^p X_{ki} \hat{\Psi}_i$.

In our particular modeling, due to the independence across individuals, it is possible for us to perform cross validation as follows. Given a number of subjects N and number of folds $K < N$, we divide the set of individuals $\{1, 2, \dots, N\}$ into K groups $\mathcal{G}_1, \dots, \mathcal{G}_K$. Then, for each $k \in [K]$, the k -th fold consist of the union of all the data corresponding to the individuals in $\mathcal{G}_k$.

While simple, this approach for cross-validation cannot be used in usual VAR models, or even the Multi-VAR and GIMME methods. The main reason is that our model assumes a structural relationship between the exogenous covariates and the individual time-series model. Therefore, the training on a set of individuals $\mathcal{G} \subset [N]$ can be used to infer the time-series coefficients for a new individual $i \in [N] \setminus \mathcal{G}$. In contrast, methods like GIMME, Multi-VAR and their extensions must train individual coefficients for each individual separately. Therefore, fitting the time-series models for a subset of individuals $\mathcal{G} \subset [N]$ cannot be exploited to determine the coefficients of a new individual $i \in [N] \setminus \mathcal{G}$, or at least not directly.

On our case, since we assume that there is an underlying additive structure that connects all the individual time-series models, it is possible to train both the individual and common structure simultaneously.

3.4.2 BIC

In addition to cross-validation, we alternatively propose the fitting of our model by selecting the tuning parameter that minimizes the Bayesian information criteria (BIC). Given $\lambda_1, \lambda_2 > 0$, we define,

$$\text{BIC}(\lambda_1, \lambda_2) = \left\| \mathbf{Z} - \mathbf{W} \widehat{\Psi}(\lambda_1, \lambda_2) \right\|_F^2 + \left\| \widehat{\Psi}(\lambda_1, \lambda_2) \right\|_0 \log(T - N). \quad (12)$$

Through numerical results, we have verified that BIC is a measure that leads to more sparse estimates of the matrix Ψ when compared to the estimate obtained through cross-

validation.

4 Theoretical Properties

In this section, we explore the theoretical properties of our proposed MOD-VAR, and provide high-dimensional consistency guarantees of estimation. In particular, we are interested in the estimation error $\|\hat{\Phi} - \Phi^*\|_F$ and ℓ_1 -error $\|\hat{\Phi} - \Phi^*\|_1$. To explore this, consider the MOD-VAR matrix of true parameters Φ^* is s -sparse: $\|\Phi^*\|_0 \leq s$. Furthermore, we assume that all time series have the same number of time observations, *i.e.*, $T := T_1 = T_2 = \dots = T_n$. Then, we denote N the total number of observations available as $N = n(T - 1)$. Consider the following set of assumptions.

Assumption 1 (Bounded moderators). *Assume the moderator vectors $\mathbf{X}, \mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are independent and identically distributed, with mean zero, and covariance matrix $\Sigma_X = \mathbb{E}[\mathbf{X}\mathbf{X}']$. Furthermore, we assume $\mathbf{X}$ bounded, *i. e.*, $|X_i| \leq 1$. Furthermore, assume that for some constant $0 < \eta < 1$, we have that $\sup_{\|\mathbf{x}\|_\infty \leq 1} \|\Phi^*(\mathbf{x})\|_2 < \eta < 1$*

Assumption 2 (Sufficient sample size). *Consider Assume that the total number of observations N satisfies $N > 16 \log(pd^2)$. Furthermore, the number of subjects n satisfies that $n > 16 \log(2n) \log(pd^2)$.*

Assumption 3 (Bounded eigenvalues). *Let $\phi_1 = \inf_{\|\mathbf{x}\|_\infty \leq 1} \Lambda_{\min}(\text{Cov}(Y_t^k | X_k))$, and $\alpha = \phi_1 \Lambda_{\min}(\Sigma_X)/2$. There exists a universal constant $C > 0$ such that, for some $t > 0$,*

$$\alpha > Cs^2 \left(\sqrt{\frac{\log(d^2p) + t}{n}} + \frac{\log(2n)(\log(pd^2) + t)}{n} \right).$$

Theorem 1. *Consider multiple subject data $\{(\mathbf{Y}_k, \mathbf{X}_k)\}_{k=1}^n$ that follows the MOD-VAR model (4). Furthermore, assume that Assumptions 1, 2 hold, and for a given $t > 0$ Assumption 3 are satisfied. Let*

$$C_M = \Lambda_{\max}(\Sigma_\varepsilon)(2 + 4\eta/(1 - \eta)^2).$$

Then, assuming that $\lambda := \lambda_1 = \lambda_2$ satisfies the inequality

$$\lambda > 16\pi C_M \sqrt{\frac{\log(pd^2)}{N}},$$

we have that, with probability at least $1 - 3e^{-t} - (6/(pd^2))^2$,

$$\|\hat{\Phi} - \Phi^*\|_2 \leq \frac{24\sqrt{s}\pi C_M}{\phi_1 \lambda_{\min}(\Sigma_X)} \sqrt{\frac{\log(pd^2)}{N}};$$

$$\|\hat{\Phi} - \Phi^*\|_1 \leq \frac{144s\pi C_M}{\phi_1 \lambda_{\min}(\Sigma_X)} \sqrt{\frac{\log(pd^2)}{N}}.$$

5 Numerical Simulations

6 Analysis of Mood and Behavior Data

7 Discussion

8 Old Intro

- The difference between multilevel-VAR, multi-VAR, and MOD-VAR lies in the specification of the idiographic structure of the time series equation. (See photo we took on 2/12.) Multilevel-VAR assumes that the "leftover" effect within is random, Multi-VAR assumes it is systematic, and MOD-VAR assumes it is systematic. ML-VAR includes a component in the idiographic structure that relates covariates to the time series (which MOD-VAR does as well), but multi-VAR does not. Because MOD-VAR includes covariates AND treats that "leftover" effect as systematic (i.e., important individual lagged structure), this distinguishes it from each of the other methods.
- There is a growing availability of longitudinal data, which allows for an increase in the breath and depth of research regarding how specific psychometric measurements influence behavior, well-being and daily progress.
- Often, this data takes the form of a set of time-series, each associated with an individual patient. When modeling, we must take into account both the temporal dependency of the data (prior psychological states are likely to influence future states), and the dependencies among the psychometric measurements.
- Many methods have been proposed for the estimation of such dependencies such as sparse vector auto-regression models [1], dynamic factor analysis [10], and the unified structural equation model (USEM) [5].
- In psychometric research, it is vital to recognize the effect that individual characteristics of each subject can have in the overall fit of the model and interpretation. From this, a variety of works have focused on the estimation of a common (or nomothetic) structure across all individuals, as well as individual (or idiographic) structure that is exclusive to each subject. The GIMME method [4] performs the estimation of the network of psychometric relationships via the estimation of multiple USEM model, with a variable selection procedure that estimates model overlaps. More recently, the MultiVAR method [3] estimates the individual and common structure through a penalized optimization approach. Further extensions to consider the potential structure of subgroups has been considered for both approaches [7; 2].

- Often, in psychometric or MRI studies, our subject-specific time series data is accompanied by additional clinical data. The incorporation of this additional information may provide further insight into behavior, but its use has been largely ignored.
- One important omission from these approaches is to explain *why* each individual has a particular idiographic structure. While the common network has a straightforward interpretation, the individual structures are not necessarily easy to interpret or process. Therefore, the output of GIMME or MultiVAR can only measure the existence of this idiographic structure, without explaining in any way the origin or motivation of potential underlying patterns.
- In the current writeup, we propose a method for estimating the nomothetic and idiographic structure of multi-subject time series, where the time dependence of each individual depends on a set of underlying covariates. This way, the idiographic effects are not simply unexplained individual structure, but are instead linked to the patients' individual characteristics. This allows for an actual interpretation of the idiographic structure.
- **Main advantages:** while other methods allow for modeling nomothetic and idiographic structure, the motivation for why such variations occur remains largely unknown. The explanation is simply: there is variation across individuals. This method connects the individual variation to underlying features, allowing us to determine the nature of the nomothetic structure, and to provide potential explanations for the origin of the idiographic structure. This can then be used as useful knowledge for further research or downstream statistical analysis.

9 Description of the Model

- Consider we have a study with N subjects. For each subject $1 \leq k \leq N$, we have access to two data modes: (i) a p -dimensional vector of covariates $\mathbf{X}_k \in \mathbb{R}^p$, and (ii) a d -dimensional time series of length T_k , given as $\mathbf{Y}_k = \{\mathbf{Y}_t^{(k)} \in \mathbb{R}^d : 1 \leq t \leq T_k\}$.
- In the usual GIMME or MultiVAR frameworks [4; 3], the aim is to model the time-

series with a common+individual decomposition, usually of the form:

$$\mathbf{Y}_t^{(k)} = \sum_{\ell=1}^q (\Psi_\ell^c + \Psi_\ell^{(k)}) \mathbf{Y}_{t-\ell}^{(k)} + \mathbf{e}_t^{(k)}. \quad (\text{MultiVAR});$$

$$\mathbf{Y}_t^{(k)} = (\mathbf{A}^c + \mathbf{A}^{(k)}) \mathbf{Y}_t^{(k)} + \sum_{\ell=1}^q (\Phi_\ell^c + \Phi_\ell^{(k)}) \mathbf{Y}_{t-\ell}^{(k)} + \varepsilon_t^{(k)}. \quad (\text{GIMME}).$$

- While these approaches are useful at extracting common and individual structure, they miss the opportunity to exploit the potentially useful additional information contained in $\{\mathbf{Y}_k\}_{k=1}^N$.
- To exploit it, we assume that the time-series model coefficients depend on the values of Y . We consider the following models:

$$\mathbf{Y}_t^{(k)} = \sum_{\ell=1}^q \Psi_\ell(\mathbf{X}_k) \mathbf{Y}_{t-\ell}^{(k)} + \mathbf{e}_t^{(k)}. \quad (\text{MultiVAR}); \quad (13)$$

$$\mathbf{Y}_t^{(k)} = \mathbf{A}(\mathbf{X}_k) \mathbf{Y}_t^{(k)} + \sum_{\ell=1}^q \Phi_\ell(\mathbf{X}_k) \mathbf{Y}_{t-\ell}^{(k)} + \varepsilon_t^{(k)}. \quad (\text{GIMME}). \quad (14)$$

Notice that, instead of being fixed, the set of coefficients $\Psi_\ell(\mathbf{X}_k), \mathbf{A}(\mathbf{X}_k), \Phi_\ell(\mathbf{X}_k)$ now depend on the individual covariate data. From this, the individual structure of each cannot be arbitrary, but it now depends on the specific features $\mathbf{X}_k$.

- As a first approach, lets assume that these parameters have an *additive* relationship to the underlying covariates. For a vector $\mathbf{x} = (x_1, x_2, \dots, x_p)^\top \in \mathbb{R}^p$

$$\Psi_\ell(\mathbf{x}) = \Psi_{\ell 0} + x_1 \Psi_{\ell 1} + x_2 \Psi_{\ell 2} + \dots + x_p \Psi_{\ell p};$$

$$\mathbf{A}(\mathbf{x}) = \mathbf{A}_0 + x_1 \mathbf{A}_1 + x_2 \mathbf{A}_2 + \dots + x_p \mathbf{A}_p;$$

$$\Phi_\ell(\mathbf{x}) = \Phi_{\ell 0} + x_1 \Phi_{\ell 1} + x_2 \Phi_{\ell 2} + \dots + x_p \Phi_{\ell p}.$$

Here, the matrices $\{\Psi_{\ell 0}\}_{\ell=1}^q, \mathbf{A}_0, \{\Phi_{\ell 0}\}_{\ell=1}^q$ represent the structure that is common (nomothetic) among all subjects. The remaining parameters are affected by $\mathbf{x}$, so they are dependent on the individual characteristics of each subject.

- **Example:** We consider a VAR(1) model, where there is a single lagged relationship, *i.e.* $q = 1$, and two subject-level covariates $p = 2$. In that case, given that $\mathbf{X}_k =$

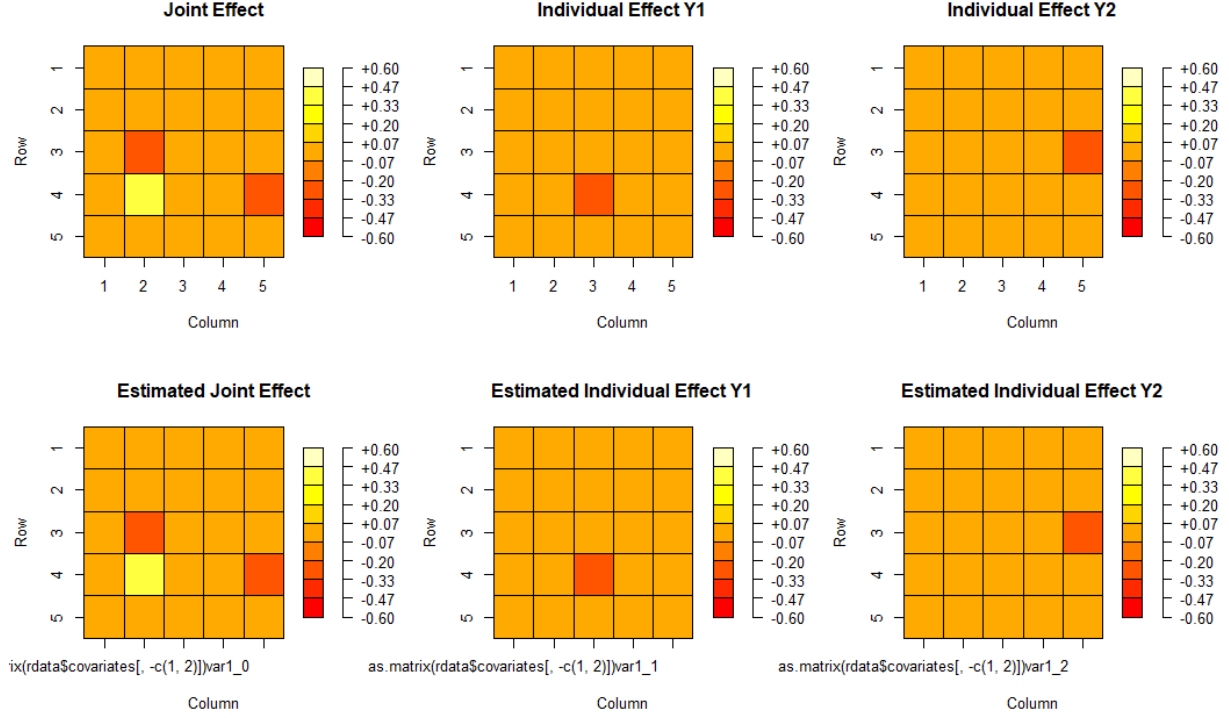


Figure 2: Illustration of the true and estimated Regression VAR parameters. For very large sample sizes, the traditional OLS method is successful. It would now be interesting to explore how regularization or model selection can help in higher dimension.

$(X_{k1}, X_{k2})^\top$ our equations simplify to:

$$\begin{aligned} \mathbf{Y}_t^{(k)} &= \Psi(\mathbf{X}_k) \mathbf{Y}_{t-1}^{(k)} + \mathbf{e}_t^{(k)} \\ &= [\Psi_0 + X_{k1} \Psi_1 + X_{k2} \Psi_2] \cdot \mathbf{Y}_{t-1}^{(k)} + \mathbf{e}_t^{(k)} \end{aligned}$$

Notice that each subject k will have their own VAR structure. This structure will incorporate a common structure shared across all subjects $\Psi_0 \in \mathbb{R}^{d \times d}$. It also has individual structure. Now, instead of having this individual structure vary arbitrarily, it has the form $Y_{k1} \Psi_1 + Y_{k2} \Psi_2$. Therefore, it depends on known information $Y_k = (Y_{k1}, Y_{k2})'$ about each subject.

- In Figure 3, we provide an example of how common and individual effects in a VAR time series model can relate to underlying subject-level covariates. Here, the dimension of our time series is $d = 5$, the number of subject-level covariates is $p = 2$, and the lag in the model is $q = 1$. We generate data for $N = 50$ subjects, all with a total of $T = 100$ time points.

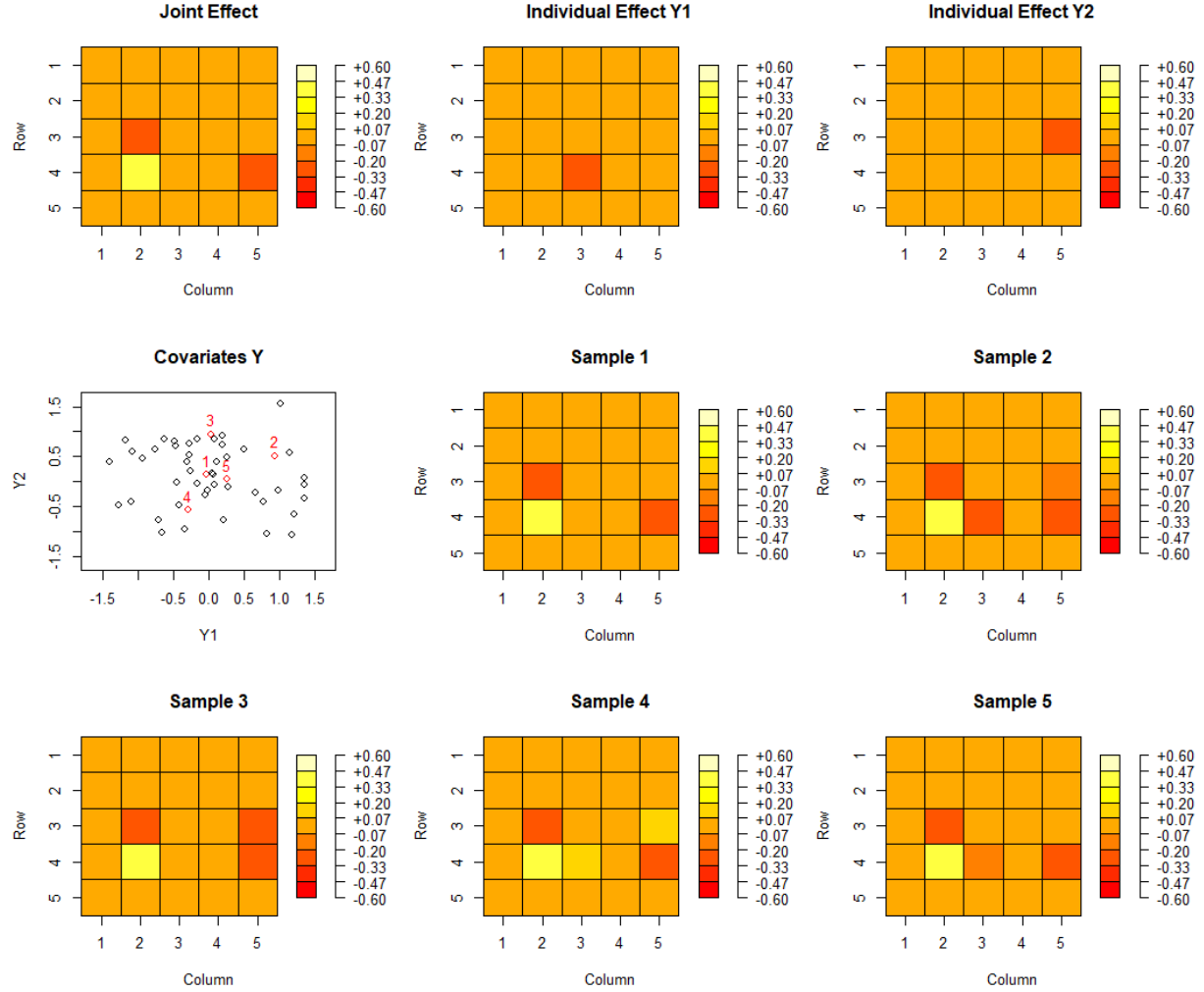


Figure 3: Illustration of the Regression VAR framework with 2 covariate-dependent effects. Top-panels: visualization of the joint effects that are common across all subjects. We also visualize the effects that are individual to variables X_1 and X_2 . Center-left panel: Visualization of the covariate information for $N = 50$ subjects. We highlight the covariate information of the first 5 subjects. Remaining panels: Visualization of the effects for the first 5 subjects. We observe that, as subject 1 has nearly zero for both X_1 and X_2 , the effects are mostly just the joint effects. Subject 3 has a high X_2 covariate, but low X_1 covariate, and therefore its effects are a mixture of the joint and X_2 effects. Subject 2 has large covariates for both X_1 and X_2 , so its time series effects are a mixture of all.

- For this same example, we performed a traditional LS fitting, to see if we could successfully recover the common/individual structure of the time series. As we see in Figure 2, our estimation for very large sample sizes is successful.

10 Fitting the RVAR Model with LASSO Penalty

In this section, we describe the fitting of the RVAR model of the form (13), with 1-lag relationships, *i.e.* $q = 1$. Assuming that, for each individual $k = 1, \dots, N$, we can capture the VAR relationships in matrix form as:

$$\underbrace{\begin{bmatrix} (\mathbf{Y}_{T_k}^k)' \\ (\mathbf{Y}_{T_k-1}^k)' \\ \vdots \\ (\mathbf{Y}_1^k)' \end{bmatrix}}_{\mathbf{Z}^k} = \underbrace{\begin{bmatrix} (\mathbf{Y}_{T_k-1}^k)' & X_{k1}(\mathbf{Y}_{T_k-1}^k)' & X_{k2}(\mathbf{Y}_{T_k-1}^k)' & \dots & X_{kp}(\mathbf{Y}_{T_k-1}^k)' \\ (\mathbf{Y}_{T_k-2}^k)' & X_{k1}(\mathbf{Y}_{T_k-2}^k)' & X_{k2}(\mathbf{Y}_{T_k-2}^k)' & \dots & X_{kp}(\mathbf{Y}_{T_k-2}^k)' \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ (\mathbf{Y}_1^k)' & X_{k1}(\mathbf{Y}_1^k)' & X_{k2}(\mathbf{Y}_1^k)' & \dots & X_{kp}(\mathbf{Y}_1^k)' \end{bmatrix}}_{\mathbf{W}^k} \underbrace{\begin{bmatrix} \Psi_0' \\ \Psi_1' \\ \vdots \\ \Psi_p' \end{bmatrix}}_{\Psi} + \underbrace{\begin{bmatrix} (\varepsilon_T^k)' \\ (\varepsilon_{T-1}^k)' \\ \vdots \\ (\varepsilon_{T-q+1}^k)' \end{bmatrix}}_{\mathbf{E}^k}.$$

This can be simplified to matrix form as

$$\mathbf{Z}^k = \mathbf{W}^k \Psi + \mathbf{E}^k \quad (15)$$

The equation (15) represents the VAR relationships for individual k . Then, we can model the behavior for all individuals via the equation

$$\underbrace{\begin{bmatrix} \mathbf{Z}^1 \\ \mathbf{Z}^2 \\ \vdots \\ \mathbf{Z}^N \end{bmatrix}}_{\mathbf{Z}} = \underbrace{\begin{bmatrix} \mathbf{W}^1 \\ \mathbf{W}^2 \\ \vdots \\ \mathbf{W}^N \end{bmatrix}}_{\mathbf{W}} \Psi + \underbrace{\begin{bmatrix} \mathbf{E}^1 \\ \mathbf{E}^2 \\ \vdots \\ \mathbf{E}^N \end{bmatrix}}_{\mathbf{E}}, \quad (16)$$

Thus, we can summarize the VAR equation model for all individuals simultaneously with the reduced matrix equation:

$$\mathbf{Z} = \mathbf{W} \Psi + \mathbf{E}, \quad (17)$$

where $\mathbf{Z} \in \mathbb{R}^{(T-N) \times d}$, $T = \sum_k T_k$, $\mathbf{W} \in \mathbb{R}^{(T-N) \times (d(p+1))}$ and $\mathbf{E} \in \mathbb{R}^{(T-N) \times d}$.

To solve for the parameter $\Psi \in \mathbb{R}^{d(p+1) \times d}$, we propose the following optimization problem,

$$\widehat{\Psi} := \widehat{\Psi}(\lambda_1, \lambda_2) = \underset{\Psi \in \mathbb{R}^{d(p+1) \times d}}{\operatorname{argmin}} \quad \|\mathbf{Z} - \mathbf{W} \Psi\|_F^2 + \lambda_1 \|\Psi_0\|_1 + \lambda_2 \|\Psi_1 \Psi_2 \dots \Psi_p\|_1, \quad (18)$$

where the hyperparameter λ_1 serves to penalize the common structure of all VAR models, and λ_2 penalizes the X covariate dependent portion of the model. We can interpret (18) as a matrix regression problem. Notice that the optimization problem (18) is equivalent to solving the separate column-wise optimization problems

$$\widehat{\Psi}^\ell(\lambda_1, \lambda_2) = \underset{\Psi^{(\ell)} \in \mathbb{R}^{d(p+1)}}{\operatorname{argmin}} \quad \|\mathbf{Z}_{\cdot \ell} - \mathbf{W} \Psi^{(\ell)}\|_F^2 + \lambda_1 \|[\Psi^{(\ell)}]_{1:d}\|_1 + \lambda_2 \|[\Psi^{(\ell)}]_{(d+1):(dp)}\|_1. \quad (19)$$

When solving the individual optimization problems (10.1), we can set the estimated parameter $\widehat{\Psi}(\lambda_1, \lambda_2) := [\widehat{\Psi}^1 \widehat{\Psi}^2 \dots \widehat{\Psi}^d] (\lambda_1, \lambda_2) \in \mathbb{R}^{d(p+1) \times d}$. Figure 2 in the previous section was derived with $\lambda_1 = \lambda_2 = 0$.

The current implementation only allows for a single lagged relationship, *i.e.* $q = 1$. We aim to create a more general implementation with lag $q > 1$ in the future. Alternatively, we can further vectorize the representation (17) into a single-response linear regression of the form

$$\text{vec}(\mathbf{Z}) = (I_d \otimes \mathbf{W}) \cdot \text{vec}(\Psi) + \text{vec}(\mathbf{E})$$

where $\text{vec}(\mathbf{A}) = [A_{11}^\top, A_{12}^\top, \dots, A_{1n}^\top]^\top \in \mathbb{R}^{ab}$ and $\mathbf{A} \otimes \mathbf{B} = [A_{ij} \cdot \mathbf{B}]_{i,j} \in \mathbb{R}^{ac \times bd}$ for any $\mathbf{A} \in \mathbb{R}^{a \times b}$ and $\mathbf{B} \in \mathbb{R}^{c \times d}$. Thus, solving (10.1) is equivalent to solving a weighted-penalty version of the LASSO [add LASSO/glmnet ref.](#)

10.1 Adaptive CD-VAR

To improve upon the performance of our original estimator, we propose a refinement of our CD-VAR estimator through an adaptive approach. For this, let $\widehat{\Psi}$ be an initial estimator of the CD-VAR model. Then, given a constant $\gamma > 0$ we define the adaptively-weighted ℓ_1 penalty as:

$$P_{ada}(\Psi) = \sum_{\ell=1}^p \sum_{i,j=1}^d |\Psi_{ij}^{(\ell)}|^{-\gamma} \cdot |\Psi_{ij}^{(w)}| \quad (20)$$

Then, for $\lambda > 0$, our adaptive CD-VAR estimator is given as a solution to the optimization problem

$$\widehat{\Psi}_{ada}(\lambda_a) = \underset{\Psi \in \mathbb{R}^{d \times (d(p+1))}}{\text{argmin}} \quad \|\mathbf{Z} - \mathbf{W}\Psi\|_F^2 + \lambda P_{ad}(\Psi).$$

In following sections, we explore how the adaptive approach can help improve the edge-recovery performance.

10.2 Tuning Parameter Selection

One of the main concerns when fitting a penalized model is to determine the best procedure for selecting tuning parameters. In particular, due to the dependence present in time series data, direct sample splitting is not feasible, leading to sophisticated procedures of data splitting to obtain a reasonable cross-validation procedure. For example [add references](#),

and what they do.

10.2.1 Cross-Validation

Contrary to other methods like GIMME and Multi-VAR, one can generate “predictive model coefficients”: with a model $\hat{\Psi} \in \mathbb{R}^{d \times (p(d+1))}$ trained on the data of N individuals $\{(\mathbf{X}_k, \mathbf{Y}^{(k)})\}_{k=1}^N$, we may provide an estimate of the coefficients of an out-of-sample individual $\{(\mathbf{X}_{N+1}, \mathbf{Y}^{(N+1)})\}$ via $\hat{\Phi}_{N+1} = \hat{\Psi}_0 + \sum_{i=1}^p X_{ki} \hat{\Psi}_i$.

In our particular modeling, due to the independence across individuals, it is possible for us to perform cross validation as follows. Given a number of subjects N and number of folds $K < N$, we divide the set of individuals $\{1, 2, \dots, N\}$ into K groups $\mathcal{G}_1, \dots, \mathcal{G}_K$. Then, for each $k \in [K]$, the k -th fold consist of the union of all the data corresponding to the individuals in $\mathcal{G}_k$.

While simple, this approach for cross-validation cannot be used in usual VAR models, or even the Multi-VAR and GIMME methods. The main reason is that our model assumes a structural relationship between the exogenous covariates and the individual time-series model. Therefore, the training on a set of individuals $\mathcal{G} \subset [N]$ can be used to infer the time-series coefficients for a new individual $i \in [N] \setminus \mathcal{G}$. In contrast, methods like GIMME, Multi-VAR and their extensions must train individual coefficients for each individual separately. Therefore, fitting the time-series models for a subset of individuals $\mathcal{G} \subset [N]$ cannot be exploited to determine the coefficients of a new individual $i \in [N] \setminus \mathcal{G}$, or at least not directly.

On our case, since we assume that there is an underlying additive structure that connects all the individual time-series models, it is possible to train both the individual and common structure simultaneously.

10.2.2 BIC

In addition to cross-validation, we alternatively propose the fitting of our model by selecting the tuning parameter that minimizes the Bayesian information criteria (BIC). Given $\lambda_1, \lambda_2 > 0$, we define,

$$\text{BIC}(\lambda_1, \lambda_2) = \left\| \mathbf{Z} - \mathbf{W} \hat{\Psi}(\lambda_1, \lambda_2) \right\|_F^2 + \left\| \hat{\Psi}(\lambda_1, \lambda_2) \right\|_0 \log(T - N). \quad (21)$$

Through numerical results, we have verified that BIC is a measure that leads to more sparse estimates of the matrix Ψ when compared to the estimate obtained through cross-validation.

11 Fitting RUSEM

The implementation of the RVAR model described in Section 10 is simply a generalization of the optimization problem considered in the Fisher et al. [3]. Notice that the RVAR focuses the estimation on the lagged effects Ψ . For the GIMME model, we are also interested in recovering contemporaneous effects. The GIMME model cannot easily be extended to the regression setting. Therefore, we need to explore a different procedure for recovering the USEM structure in the regression setting. This will require thought and research...

I explored the original GIMME reference Gates and Molenaar [4], which points towards the modification index described in Sörbom [9]. I am still trying to understand how the GIMME adapts the estimation of USEM described in Sörbom [9] to scenarios with time dependence. This will be the next task.

12 Simulation Study for R-VAR

To explore the performance of our proposed RVAR model, and to compare to other methods found in the literature, we perform a numerical simulation study.

12.1 Simulation Setup

We consider multivariate time series over $d = 10, 20, 30$ variables of interest. We consider scenarios with $N = 10, 20$ subjects, measured over $T = 30, 50, 100$ time points. For each subject $k \in \{1, 2, \dots, N\}$, we simulate a vector of $p = 2, 5$ time-invariant covariates according to the mixture distribution $\mathbf{X}_k = (X_{k1}, X_{k2}, \dots, X_{kp}) \stackrel{i.i.d.}{\sim} 0.25 \cdot \delta_0 + 0.75 \cdot \mathbf{Uniform}([-1, -0.5] \cup [0.5, 1])$. Then, we simulate a lag-1 time series dataset for the subject $k \in \{1, 2, \dots, N\}$ according to the covariate-dependent VAR model (13) with parameter $\Psi(\mathbf{X}_i)$, where

$$\Psi(\mathbf{x}) = \Psi_0 + x_1\Psi_1 + x_2\Psi_2 + \dots + x_p\Psi_p.$$

Finally, the common effect matrix $\Psi_0 \in \mathbb{R}^{d \times d}$ and the matrix of covariate-dependent effects $\Psi_1, \dots, \Psi_p \in \mathbb{R}^{d \times d}$ are generated as follows. First, we select the proportion of common non-zero entries η_c , a proportion of covariate-dependent non-zero entries η_i . We generate Ψ_0 by selecting at random $d^2 \eta_c$ entries to follow the distribution $\mathbf{Uniform}([-0.9, -0.3] \cup [0.3, 0.9])$. To generate Ψ_k , we select $d^2 \eta_i$ to follow $\mathbf{Uniform}([-0.9, -0.3] \cup [0.3, 0.9])$. In our simulations, we consider $(\eta_c, \eta_i) = \{(0.075, 0.025), (0.025, 0.075)\}$. Finally, we generate time series for the k -th subject $\{\mathbf{Y}_1^{(k)}, \mathbf{Y}_2^{(k)}, \dots, \mathbf{Y}_{T+400}^{(k)}\}$ via the model (13), with $\mathbf{e}_1^{(k)}, \mathbf{e}_2^{(k)}, \dots, \mathbf{e}_T^{(k)} \stackrel{i.i.d.}{\sim} N_d(\mathbf{0}, (0.1)\mathbf{I}_d)$, and discard the first 400 time steps.

We compare our proposed sparse CD-VAR method with methods of multi-subject vector auto-regressive models found in the literature. In particular, we consider the estimation of the time series via the individual VAR model [1], as well as the Multi-VAR method in their original and adaptive versions [3]. For the selection of the tuning parameters of the CD-VAR method, we consider training based on cross-validation, as well as based on our proposed BIC measure. Furthermore, we fit our method with both a direct and two-step adaptive approach, as described in Section 10.1.

We compare the methods in terms of several measurements of performance. First, we consider the mean root aggregated error (MRAE), and mean absolute aggregated error (MAAE), given by,

$$\begin{aligned} MRAE &= \frac{1}{N} \sum_{k=1}^N \sum_{\ell=1}^p \left\| \hat{\Psi}_\ell^k - \Psi_\ell^k \right\|_F; \\ MAAE &= \frac{1}{N} \sum_{k=1}^N \sum_{\ell=1}^p \left\| \hat{\Psi}_\ell^k - \Psi_\ell^k \right\|_1; \end{aligned}$$

12.2 Simulation Results

The results of the MRAE for $d = 30$ are presented in Figure 4. The best performance for all scenarios is obtained by the adaptive CV-SCDVAR model, closely followed by the original CV-SCDVAR model and the adaptive BIC-SCDVAR model. This findings are consistent across choices of number of explanatory covariates $p = 2, 5$, number of subjects $N = 10, 20$, common entries $p_c = 0.025, 0.075$ and number of available time points T .

13 Studying Consistency of VAR

In order to show the consistency of our CD-VAR procedure, we must first explore and understand the consistency of the plain penalized VAR model. For this, we consider a sequence $\{\mathbf{X}_t\}_{t \in \mathbb{Z}} \in \mathbb{R}^d$ distributed as a VAR(q) model of the form:

$$\mathbf{X}_t = \sum_{\ell=1}^q \Psi_\ell \mathbf{X}_{t-\ell} + \varepsilon_t, \quad (22)$$

where $\Psi_1, \Psi_2, \dots, \Psi_q \in \mathbb{R}^{d \times d}$ are the coefficient matrices. We first define the polynomial $\mathcal{A}(z) = \mathbf{I}_d - \sum_{\ell} \Psi_\ell z^\ell$ for any $z \in \mathbb{C}$. To satisfy the stability condition, we need that $\det(\mathcal{A}(z)) \neq 0$ for all $z \in \mathbb{C}$ with $|z| \leq 1$.

We define the auto-correlation matrices $\Gamma_X(i) = \text{cov}(X_{t-i}, X_t)$. Furthermore, we define the spectral density of the process as:

$$f_X(\theta) = \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \Gamma_X(\ell) e^{-i\ell\theta}, \quad \forall \theta \in [-\pi, \pi]. \quad (23)$$

Finally, we define the maximum spectral eigenvalue as,

$$\mathcal{M}(f_X) = \sup_{\theta \in [-\pi, \pi]} \Lambda_{\max}(f_X(\theta)) < \infty. \quad (24)$$

Furthermore, for any k we can define the sub-variable spectral density radius as,

$$\mathcal{M}(f_X, k) = \max_{\mathcal{J} \subset [d]; |\mathcal{J}|=k} \mathcal{M}(f_{X_{\mathcal{J}}}) \quad (25)$$

Similar to the spectral density radius, we can define the minimum spectral density radius as:

$$m(f_X) = \inf_{\theta \in [-\pi, \pi]} \Lambda_{\min}(f_X(\theta)).$$

There are several important observations:

- The existence of the spectral density guarantees that

$$\Gamma_X(\ell) = \int_{-\pi}^{\pi} f_X(\theta) e^{i\ell\theta} d\theta, \quad \forall \ell \in \mathbb{Z}. \quad (26)$$

- The spectral density radius $\mathcal{M}(f_X)$ provides a measurement of the degree of dependence present in the data. If there is low dependence, then each of the terms $\|\Gamma_X(\ell)\|$ will decay fast for large ℓ , so the eigenvalue $\Lambda_{\max}(f_X(\theta))$ will be small. If there is high dependence, then the terms $\|\Gamma_X(\ell)\|$ will stay large and the eigenvalues $\Lambda_{\max}(f_X(\theta))$ will remain large.

- We have that:

$$\mathcal{M}(f_X, 1) \leq \mathcal{M}(f_X, 1) \leq \dots \leq \mathcal{M}(f_X, d) = \mathcal{M}(f_X).$$

Lets write down some useful facts.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]' \in \mathbb{R}^{n \times d}$ be the matrix of observations from our time series, and $\text{vec}(\mathbf{X}) = [\mathbf{X}_1', \mathbf{X}_2', \dots, \mathbf{X}_n']' \in \mathbb{R}^{nd}$. Let $\Upsilon_n^X = \text{cov}(\text{vec}(\mathbf{X})) \in \mathbb{R}^{nd \times nd}$. In particular, we have that

$$\Upsilon_n^X = \begin{pmatrix} \Gamma_X(0) & \Gamma_X(1) & \dots & \Gamma_X(n) \\ \Gamma_X(-1) & \Gamma_X(0) & \dots & \Gamma_X(n-1) \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_X(-n) & \Gamma_X(-n+1) & \dots & \Gamma_X(0) \end{pmatrix} \in \mathbb{R}^{nd \times nd}. \quad (27)$$

First, we can show that the spectra of the covariance Υ_n^X is bounded by the spectral density.

Proposition 1 (Proposition 2.3 of ??). *For any $n \geq 1$ and $d \geq 1$,*

$$2\pi m(f_X) \leq \Gamma_{\min}(\Upsilon_n^X) \leq \Gamma_{\max}(\Upsilon_n^X) \leq 2\pi \mathcal{M}(f_X);$$

and in particular,

$$2\pi m(f_X) \leq \Gamma_{\min}(\Lambda_X(0)) \leq \Gamma_{\max}(\Lambda_X(0)) \leq 2\pi \mathcal{M}(f_X).$$

Proof. Observe that for any $\mathbf{v} \in \mathbb{R}^{nd}$ with $\|\mathbf{v}\| = 1$, and $\mathbf{v} = [\mathbf{v}_1', \dots, \mathbf{v}_n']' \in \mathbb{R}^{nd}$, we have that for $G(\theta) = \sum \mathbf{v}_i e^{-ir\theta}$,

$$\int_{-\pi}^{\pi} G^*(\theta) G(\theta) d\theta = \sum_{r,s} \int_{-\pi}^{\pi} v_s' v_r e^{i(r-s)\theta} d\theta = \sum_r 2\pi \|\mathbf{v}_r\|_2^2 = 2\pi$$

On the other hand,

$$\begin{aligned} \mathbf{v}' \Upsilon_n^X \mathbf{v} &= \sum_{r,s} \mathbf{v}_s' \Gamma_X(s-r) \mathbf{v}_r \\ &= \sum_{s,r} \int_{-\pi}^{\pi} \mathbf{v}_s' f_X(\theta) \mathbf{v}_r e^{i(s-r)\theta} d\theta \\ &= \int_{-\pi}^{\pi} G^*(\theta) f_X(\theta) G(\theta) d\theta. \end{aligned}$$

Finally, notice that for any $\theta \in [-\pi, \pi]$

$$m(f_X) G^*(\theta) G(\theta) \leq G^*(\theta) f_X(\theta) G(\theta) \leq \mathcal{M}(f_X) G^*(\theta) G(\theta),$$

$$\therefore 2\pi \cdot m(f_X) \leq \Lambda_{\min}(\Upsilon_n^X) \leq \Lambda_{\max}(\Upsilon_n^X) \leq 2\pi \cdot \mathcal{M}(f_X).$$

□

Furthermore, we consider the following sub-matrix concentration inequality. Let $S = \mathbf{X}'\mathbf{X}/n$. We want to study how the matrix S concentrates around $\Gamma_X(0) \in \mathbb{R}^{d \times d}$.

Proposition 2 (Proposition 2.4 of ??). *For a stationary centered process $\mathbf{X}$ and k -sparse vectors $\mathbf{v}, \mathbf{u} \in \mathbb{R}^d$ with $\|\mathbf{v}\| = \|\mathbf{u}\| = 1$,*

$$\mathbb{P}[|\mathbf{v}'(S - \Gamma_X(0))\mathbf{v}| > 2\pi\mathcal{M}(f_X, k)\eta] \leq 2 \exp\{-cn \min\{\eta^2, \eta\}\}, \quad (28)$$

$$\mathbb{P}[|\mathbf{u}'(S - \Gamma_X(0))\mathbf{v}| > 2\pi\mathcal{M}(f_X, 2k)\eta] \leq 6 \exp\{-cn \min\{\eta^2, \eta\}\}, \quad (29)$$

$$\mathbb{P}[|S_{ij} - [\Gamma_X(0)]_{ij}| > 2\pi\mathcal{M}(f_X, 2)\eta] \leq 6 \exp\{-cn \min\{\eta^2, \eta\}\} \quad (30)$$

Proof. Lets just show (28) first. Let $\mathcal{J}$ be the support of $\mathbf{v}$. Observe that $\text{vec}(\mathbf{X}) \sim N_d(\mathbf{0}, \Upsilon_n^X)$. Then $\text{vec}(\mathbf{X}_J) \sim N_d(\mathbf{0}, \Upsilon_n^{X_J})$. From this $Y = \mathbf{X}\mathbf{v} = \mathbf{X}_J\mathbf{v}_J \in \mathbb{R}^n$. Then, we have

$$\text{cov}(Y_r, Y_s) = \text{cov}([\mathbf{X}_J\mathbf{v}_J]_r, [\mathbf{X}_J\mathbf{v}_J]_s) = \text{cov}(\mathbf{X}_r'\mathbf{v}, \mathbf{X}_s'\mathbf{v}) = \mathbf{v}'\text{cov}(\mathbf{X}_r, \mathbf{X}_s)\mathbf{v} = \mathbf{v}'\Gamma_X(r-s)\mathbf{v}.$$

Thus, $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{Q})$ with $Q_{rs} = \mathbf{v}'\Gamma_X(r-s)\mathbf{v}$. Then, with $\mathbf{v}'S\mathbf{v} = \frac{1}{n}\mathbf{Y}'\mathbf{Y} = \frac{1}{n}\mathbf{Z}'\mathbf{Q}\mathbf{Z}$ where $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I}_n)$. By the Hanson-Wright inequality, (Rudelson & Vershynin, 2013) we have that:

$$\begin{aligned} \mathbb{P}[|\mathbf{v}'(S - \Gamma_X(0))\mathbf{v}| \geq \zeta] &= \mathbb{P}[|\mathbf{Z}\mathbf{Q}\mathbf{Z} - \mathbb{E}[\mathbf{Z}\mathbf{Q}\mathbf{Z}]| \geq n\zeta] \\ &\leq 2 \exp\left\{-cn \min\left[\frac{n^2\zeta^2}{\|\mathbf{Q}\|_F^2}, \frac{n\zeta}{\|\mathbf{Q}\|_2}\right]\right\}. \end{aligned}$$

Since $\|\mathbf{Q}\|_F^2/n \leq \|\mathbf{Q}\|_2^2$, and setting $\zeta = \|\mathbf{Q}\|_2\eta$, we have:

$$\mathbb{P}[|\mathbf{v}'(S - \Gamma_X(0))\mathbf{v}| \geq \eta\|\mathbf{Q}\|_2] \leq 2 \exp\{-cn \min[\eta^2, \eta]\}. \quad (31)$$

Finally, we observe that for $\mathbf{w} \in \mathbb{R}^n$, and $\mathbf{W} = [w_1\mathbf{v}'_J, w_2\mathbf{v}'_J, \dots, w_n\mathbf{v}'_J]'$

$$\begin{aligned} \mathbf{w}'\mathbf{Q}\mathbf{w} &= \sum_{s,r} \mathbf{w}_s \mathbf{w}_r Q_{s,r} = \sum_{s,r} \mathbf{w}_s \mathbf{w}_r \mathbf{v}'_J \Gamma_{X_J}(r-s) \mathbf{v}_J \\ &= \mathbf{W}' \Upsilon_n^{X_J} \mathbf{W} \leq \Lambda_{\max}(\Upsilon_n^{X_J}) \leq 2\pi\mathcal{M}(f_X, k) \end{aligned}$$

Thus, we can substitute $\|\mathbf{Q}\|_2$ by $2\pi\mathcal{M}(f_X, k)$ in (31). $\square$

14 Consistency of MOD-VAR

14.1 Setting and Assumptions

Consider observations on n subjects, with vector of p moderators $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \in \mathbb{R}^p$ random vectors with $\mathbb{E}[\mathbf{X}_k] = \mathbf{0}$ and entrywise bounded support: $|X_{ki}| \leq B$. Let $\Psi_0^*, \Psi_1^*, \dots, \Psi_p^* \in \mathbb{R}^{d \times d}$ be the true model parameters, and we define $\Psi^* : \mathbb{R}^p \rightarrow \mathbb{R}^{d \times d}$ via the mapping $\Psi^*(\mathbf{x}) = \Psi_0 + \sum_{i=1}^p x_i \Psi_i$. For each subject $k \in [n]$, we have T measurements of a d -dimensional time-series $\{\mathbf{Y}_t : 1 \leq t \leq T\}$, with the distribution:

$$\mathbf{Y}_t^k = \Psi^*(\mathbf{X}_k) \mathbf{Y}_{t-1}^k + \boldsymbol{\varepsilon}_t^k, \quad \forall 2 \leq t \leq T, \quad (32)$$

where $\boldsymbol{\varepsilon}_2^k, \boldsymbol{\varepsilon}_3^k, \dots, \boldsymbol{\varepsilon}_T^k \stackrel{i.i.d.}{\sim} N_d(\mathbf{0}, \Sigma_\varepsilon)$. We denote:

$$\mathbf{Z}^k = \begin{pmatrix} (\mathbf{Y}_T^k)' \\ (\mathbf{Y}_{T-1}^k)' \\ \vdots \\ (\mathbf{Y}_2^k)' \end{pmatrix}, \quad \mathbf{Y}^k = \begin{pmatrix} (\mathbf{Y}_{T-1}^k)' \\ (\mathbf{Y}_{T-2}^k)' \\ \vdots \\ (\mathbf{Y}_1^k)' \end{pmatrix}, \quad \mathbf{E}^k = \begin{pmatrix} (\boldsymbol{\varepsilon}_T^k)' \\ (\boldsymbol{\varepsilon}_{T-1}^k)' \\ \vdots \\ (\boldsymbol{\varepsilon}_2^k)' \end{pmatrix}. \quad (33)$$

Thus, we can summarize the autoregressive structure for each subject given by (32) via the matrix form equation:

$$\mathbf{Z}^k = \mathbf{Y}^k \Psi^*(\mathbf{X}_k) + \mathbf{E}^k.$$

by a slight abuse of notation, we assume that for all $k \in [n]$, the vector of moderators $\mathbf{X}_k \in \mathbb{R}^p$ contains a zero indexed entry $X_{k0} = 1$. We define $\Psi^* := [\Psi_0, \Psi_1, \dots, \Psi_p]' \in \mathbb{R}^{(pd) \times d}$, and we define

$$\mathbf{W}^k = [\mathbf{Y}^k, X_{k1} \mathbf{Y}^k, \dots, X_{kp} \mathbf{Y}^k] = \mathbf{X}_k' \otimes \mathbf{Y}^k \in \mathbb{R}^{(T-1) \times (pd)}.$$

Then, one can show that $\mathbf{Y}^k \Psi^*(\mathbf{X}_k) = \mathbf{W}^k \Psi^*$. Now, we define

$$\mathbf{Z} = \begin{pmatrix} \mathbf{Z}^1 \\ \mathbf{Z}^2 \\ \vdots \\ \mathbf{Z}^n \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} \mathbf{W}^1 \\ \mathbf{W}^2 \\ \vdots \\ \mathbf{W}^n \end{pmatrix}, \quad \mathbf{E} = \begin{pmatrix} \mathbf{E}^1 \\ \mathbf{E}^2 \\ \vdots \\ \mathbf{E}^n \end{pmatrix}. \quad (34)$$

Then, we can resume the entire model to our simplified matrix notation $\mathbf{Z} = \mathbf{W} \Psi^* + \mathbf{E}$.

For a matrix $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m] \in \mathbb{R}^{k \times m}$ with column vectors $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^k$, we

denote $\mathbf{vec}(\mathbf{A}) = (\mathbf{a}'_1, \mathbf{a}'_2, \dots, \mathbf{a}'_m)' \in \mathbb{R}^{km}$. Notice that, for any $\Psi \in \mathbb{R}^{(pd) \times d}$,

$$\begin{aligned} \|\mathbf{Z} - \mathbf{W}\Psi^*\|_F^2 &= \|\mathbf{vec}(\mathbf{Z}) - \mathbf{vec}(\mathbf{W}\Psi)\|_2^2 = \|\mathbf{vec}(\mathbf{Z}) - (\mathbf{I}_d \otimes \mathbf{W})\mathbf{vec}(\Psi)\|_2^2 \\ &= \mathbf{vec}(\mathbf{Z})'\mathbf{vec}(\mathbf{Z}) - 2\mathbf{vec}(\mathbf{Z})'(\mathbf{I}_d \otimes \mathbf{W})\mathbf{vec}(\Psi) + \mathbf{vec}(\Psi)'(\mathbf{I}_d \otimes \mathbf{W}'\mathbf{W})\mathbf{vec}(\Psi). \end{aligned}$$

To simplify our notation, we define $\beta = \mathbf{vec}(\Psi)$, $\mathbf{y} = \mathbf{vec}(\mathbf{Z})$, $\gamma = \frac{1}{N}(\mathbf{I}_d \otimes \mathbf{W}')\mathbf{y}$ and $\hat{\Gamma} = \frac{1}{N}(\mathbf{I}_d \otimes \mathbf{W}'\mathbf{W})$. In particular, we denote $\beta^* = \mathbf{vec}(\Psi^*)$. Finally, we denote $N = (T-1)n$ and $q = pd^2$. Then, assuming that $\lambda_2 = 0$, we can denote our proposed optimization problem:

$$\hat{\Psi} = \underset{\Psi \in \mathbb{R}^{(pd) \times d}}{\operatorname{argmin}} \frac{1}{N} \|\mathbf{vec}(\mathbf{Z}) - \mathbf{vec}(\mathbf{W}\Psi)\|_F^2 + \|\Psi\|_1 \quad (35)$$

into the equivalent form

$$\hat{\beta} = \underset{\beta \in \mathbb{R}^{pd^2}}{\operatorname{argmin}} -2\beta'\gamma + \beta'\hat{\Gamma}\beta + \lambda\|\beta\|_1. \quad (36)$$

First, we consider the following set of high-level assumptions.

Definition 1 (Restricted Eigenvalue). We say symmetric matrix $\Gamma \in \mathbb{R}^{q \times q}$ satisfies the restricted eigenvalue condition with parameters $\alpha, \tau > 0$ if for all $\theta \in \mathbb{R}^q$,

$$\theta'\hat{\Gamma}\theta \geq \alpha\|\theta\|_2^2 - \tau\|\theta\|_1^2. \quad (37)$$

We denote that a matrix $\Gamma \in \mathbb{R}^{q \times q}$ satisfies (37) as $\Gamma \in \text{RE}(\alpha, \tau)$.

Assumption 4. *There exists a function $Q(\beta^*, \Sigma_\varepsilon, B) > 0$ such that,*

$$\|\gamma - \hat{\Gamma}\beta^*\|_\infty \leq Q(\beta^*, \Sigma_\varepsilon, B) \sqrt{\frac{2\log(d) + \log(p)}{N}}.$$

Definition 2 (Conditional spectral Density). Assume $(\mathbf{X}_k, \{\mathbf{Y}_t^k \in \mathbb{R}^d\}_{t=1}^T)$ is a moderator vector autoregressive model, where $\{\mathbf{Y}_t^k \in \mathbb{R}^d\}_{t=1}^T$ follows (32). We define the following.

- We define the *conditional autocovariance function* as $\Gamma_{Y^k|X_k}(h) = \mathbf{Cov}(\mathbf{Y}_t^k, \mathbf{Y}_{t+h}^k | \mathbf{X}_k)$.
- We define the *conditional spectral density function* as

$$f_{Y^k|X_k}(\theta) := \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \Gamma_{Y^k|X_k}(\ell) e^{-i\ell\theta}, \quad \theta \in [-\pi, \pi].$$

- We denote the maximum spectral eigenvalue by

$$\mathcal{M}(f_{Y^k|X_k}) := \sup_{\theta \in [-\pi, \pi]} \Lambda_{\max}(f_{Y^k|X_k}(\theta)).$$

- For any $\mathbf{x} \in \mathbb{R}^p$, consider the matrix polynomial $\mathcal{A}_{\mathbf{x}}(z) = \mathbf{I}_d - \Psi^*(\mathbf{x})z$, for $z \in \mathbb{C}$.

Then, we define the quantities:

$$\mu_{\min}(\mathcal{A}_{\mathbf{x}}) := \min_{|z|=1} \Lambda_{\min}(\mathcal{A}_{\mathbf{x}}^*(z)\mathcal{A}_{\mathbf{x}}(z)); \quad (38)$$

$$\mu_{\max}(\mathcal{A}_{\mathbf{x}}) := \max_{|z|=1} \Lambda_{\max}(\mathcal{A}_{\mathbf{x}}^*(z)\mathcal{A}_{\mathbf{x}}(z)). \quad (39)$$

Definition 3 (α -Orlicz norms). For a random variable $X \in \mathbb{R}$ and $\alpha > 0$, we define the α -Orlicz norm of X as

$$\|X\|_{\psi_{\alpha}} := \inf\{c > 0 : \mathbb{E}[\exp\{|X|^{\alpha}/c^{\alpha}\}] \leq 2\}.$$

14.2 Useful Lemmas

Lemma 1 (Lemma 12 in Loh and Wainwright [8]). *Let $\Gamma \in \mathbb{R}^{q \times q}$, $s \geq 1$ and $\delta > 0$. Suppose Γ satisfies the deviation bound*

$$\sup_{\|\mathbf{v}\|_2=1, \|\mathbf{v}\|_0 \leq 2s} |\mathbf{v}'\Gamma\mathbf{v}| < \delta.$$

Then, for all $\mathbf{v} \in \mathbb{R}^q$

$$|\mathbf{v}'\Gamma\mathbf{v}| \leq 27\delta(\|\mathbf{v}\|_2^2 + \frac{1}{s}\|\mathbf{v}\|_1^2).$$

Lemma 2 (Lemma B.1 of Basu and Michailidis [1]). *If $\mathbf{W}'\mathbf{W}/N \in \text{RE}(\alpha, \tau)$, then the matrix $I_d \otimes \mathbf{W}'\mathbf{W}/N$ satisfies $I_d \otimes \mathbf{W}'\mathbf{W}/N \in \text{RE}(\alpha, \tau)$ as well.*

Lemma 3 (Proposition 2.4 of Basu and Michailidis [1]). *Consider a centered Gaussian time series $\{Y_t\}_{t \in \mathbb{Z}}$ with noise matrix $\mathbf{E} \in \mathbb{R}^{(T-1) \times d}$ such that $\mathcal{M}(f_Y) < \infty$. Let $\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T]' \in \mathbb{R}^{T \times d}$ and $\mathbf{E} = [\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \dots, \boldsymbol{\varepsilon}_T]' \in \mathbb{R}^{T \times d}$. There exists a constant $c > 0$ such that, for any s -sparse vectors $\mathbf{v}, \mathbf{u} \in \mathbb{R}^p$ with $\|\mathbf{v}\|_2, \|\mathbf{u}\|_2 \leq 1$,*

$$\mathbb{P} \left[|\mathbf{u}'(\mathbf{Y}'\mathbf{E}/T)\mathbf{v}| > 2\pi\Lambda_{\max}(\Sigma_{\varepsilon}) \left(1 + \frac{1 + \mu_{\max}(\mathcal{A})}{\mu_{\min}(\mathcal{A})} \right) \eta \right] \leq 6 \exp \left[-cn \min(\eta, \eta^2) \right].$$

Lemma 4 (Exercises 2.24 and 2.44 of Vershynin [11]). *Let $X \in \mathbb{R}$ a random variable with $\|X\|_{\psi_{\alpha}} < \infty$. Then, there exists a constant $c > 0$ such that,*

- $\|X - \mathbb{E}X\|_{\psi_{\alpha}} \leq c\|X\|_{\psi_{\alpha}}$; and,
- if $|X| \leq B$, then $\|X\|_{\psi_{\alpha}} \leq cB$.

Lemma 5 (Theorem 3.4 of Kuchibhotla and Chakraborty [6]). *Suppose that $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \in \mathbb{R}^q$ are independent, mean-zero vectors such that, for some $\alpha > 0$ and $K_{n,q}, \eta_{n,q} > 0$,*

$$\max_{1 \leq i \leq n} \max_{1 \leq j \leq q} \|X_{ij}\|_{\psi_{\alpha}} \leq K_{n,q}, \quad \max_{1 \leq j \leq q} \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_{ij}^2] \leq \eta_{n,q}. \quad (40)$$

Then, for any $t > 0$, with probability at least $1 - 3e^{-t}$,

$$\left\| \frac{1}{n} \sum_{i=1}^n X_i \right\|_{\infty} \leq 7 \sqrt{\frac{\eta_{n,q}(\log(q) + t)}{n}} + \frac{C_{\alpha} K_{n,q} (\log(2n))^{1/\alpha} (\log(q) + t)^{1/\alpha^*}}{n}, \quad (41)$$

where $\alpha^* := \min\{1, \alpha\}$ and $C_{\alpha} > 0$ is some constant depending only on α .

Lemma 6 (Proposition D.2 of Kuchibhotla and Chakraborty [6]). *Let $W_1, W_2, \dots, W_k$ are possibly dependent random variables satisfying $\|W_i\|_{\psi_{\alpha_i}} < \infty$ for some $\alpha_i > 0$. Then,*

$$\left\| \prod_{i=1}^k W_i \right\|_{\psi_{\beta}} \leq \prod_{i=1}^k \|W_i\|_{\psi_{\alpha_i}}, \quad \text{where } \frac{1}{\beta} := \sum_{i=1}^k \frac{1}{\alpha_i}. \quad (42)$$

14.3 Proof of Theorem 2

Proposition 3. *Consider the MOD-VAR estimator $\hat{\beta}$ defined by (36). Assume that $\hat{\Gamma} \in \text{RE}(\alpha, \tau)$ for $s\tau < \alpha/32$. Furthermore, suppose Assumption 4 holds. Finally, assume that*

$$\lambda \geq 4Q(\beta^*, \Sigma_{\varepsilon}, B) \sqrt{\frac{2\log(d) + \log(p)}{N}}.$$

Then,

$$\|\hat{\beta} - \beta^*\|_2 \leq \frac{3\sqrt{s}\lambda}{\alpha}; \quad \|\hat{\beta} - \beta^*\|_1 \leq \frac{12s\lambda}{\alpha}.$$

Proof. If $\hat{\beta}$ is the minimizer of (36),

$$-2\hat{\beta}'\gamma + \hat{\beta}'\hat{\Gamma}\hat{\beta} + \lambda\|\hat{\beta}\|_1 \leq -2(\beta^*)'\gamma + (\beta^*)'\hat{\Gamma}\beta^* + \lambda\|\beta^*\|_1.$$

By denoting $\Delta = \hat{\beta} - \beta^*$, by basic manipulation and the triangle inequality,

$$0 \leq \Delta'\hat{\Gamma}\Delta \leq 2\Delta'(\gamma - \hat{\Gamma}\beta^*) + \lambda\|\Delta_S\|_1 - \lambda\|\Delta_{S^c}\|_1. \quad (43)$$

By Assumption 4 and Hölder's inequality,

$$2\Delta'(\gamma - \hat{\Gamma}\beta^*) \leq 2Q(\beta^*, \Sigma_{\varepsilon}, B) \sqrt{\frac{2\log(d) + \log(p)}{N}} \|\Delta\|_1 \leq \frac{\lambda}{2} \|\Delta\|_1. \quad (44)$$

Combining (43) and (44), we have the following two conclusions

$$0 \leq \frac{1}{\lambda} \Delta'\hat{\Gamma}\Delta \leq \frac{3}{2} \|\Delta_S\|_1 - \frac{1}{2} \|\Delta_{S^c}\|_1 \leq \frac{3}{2} \|\Delta_S\|_1; \quad \text{and,} \quad \|\Delta\|_1 \leq 4\|\Delta_S\|_1.$$

Note that $\|\Delta\|_1 \leq 4\|\Delta_S\|_1 \leq 4\sqrt{s}\|\Delta_S\|_2 \leq 4\sqrt{s}\|\Delta\|_2$. Since $\hat{\Gamma} \in \text{RE}(\alpha, \tau)$, we have that,

$$\Delta'\hat{\Gamma}\Delta \geq \alpha\|\Delta\|_2^2 - \tau\|\Delta\|_1^2 \geq \alpha\|\Delta\|_2^2 - \frac{\alpha}{2}\|\Delta\|_2^2 = \frac{\alpha}{2}\|\Delta\|_2^2.$$

Then,

$$\frac{\alpha}{2} \|\Delta\|_2^2 \leq \Delta'\hat{\Gamma}\Delta \leq \frac{3\lambda}{2} \|\Delta_S\|_1 \leq \frac{3\sqrt{s}\lambda}{2} \|\Delta\|_2.$$

Thus, we conclude $\|\hat{\beta} - \beta^*\|_2 \leq 3\sqrt{s}\lambda/\alpha$. Furthermore, $\|\hat{\beta} - \beta^*\|_1 \leq 4\sqrt{s}\|\Delta\|_2 \leq 12s\lambda/\alpha$. □

Proposition 4. Assume that $N > 16 \log(pd^2)$, and

$$C_M := \sup_{\mathbf{x} \in \mathbb{R}^p} \Lambda_{\max}(\Sigma_{\varepsilon}) \left(1 + \frac{1 + \mu_{\max}(\mathcal{A}_x)}{\mu_{\min}(\mathcal{A}_x)} \right) < \infty. \quad (45)$$

Then, we have that,

$$\mathbb{P} \left[\|\boldsymbol{\gamma} - \hat{\Gamma} \boldsymbol{\beta}^*\|_{\infty} \geq 4\pi C_M B \sqrt{\frac{2 \log(d) + \log(p)}{N}} \right] \leq 6 \exp\{-3c(2 \log(d) + \log(p))\}.$$

Proof. First, observe that,

$$\begin{aligned} \boldsymbol{\gamma} &= \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}') \text{vec}(\mathbf{Z}) = \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}' \mathbf{W}) \text{vec}(\Psi^*) + \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}') \text{vec}(\mathbf{E}); \\ \hat{\Gamma} \boldsymbol{\beta}^* &= \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}' \mathbf{W}) \text{vec}(\Psi^*). \end{aligned}$$

Thus, observe that,

$$\|\boldsymbol{\gamma} - \hat{\Gamma} \boldsymbol{\beta}^*\|_{\infty} = \|\text{vec}(\mathbf{W}' \mathbf{E})/N\|_{\infty} = \|\mathbf{W}' \mathbf{E}/N\|_{\max} = \max_{\substack{1 \leq i \leq p; \\ 1 \leq m, \ell \leq d}} \left| \frac{1}{n} \sum_{k=1}^n \frac{1}{T-1} X_{ki} (\mathbf{Y}_{\cdot m}^k)' \mathbf{E}_{\cdot \ell}^k \right|.$$

By Lemma 3, there exists $c > 1$ such that for any $\eta > 0$,

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{k=1}^n \frac{1}{T-1} X_{ki} (\mathbf{Y}_{\cdot m}^k)' \mathbf{E}_{\cdot \ell}^k \right| \geq 2\pi C_M \eta \right] \leq 6 \exp \left\{ -c(T-1) \min(\eta, \eta^2) \right\}.$$

Recall that $|X_{ki}| \leq B$. Using this fact, and aggregating all the independent sub-exponential bounds,

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{k=1}^n \frac{1}{T-1} X_{ki} (\mathbf{Y}_{\cdot m}^k)' \mathbf{E}_{\cdot \ell}^k \right| \geq t \right] \leq 6 \exp \left\{ -\min \left[\frac{cNt^2}{4\pi^2 C_M^2 B^2}, \frac{cNt}{2\pi C_M B} \right] \right\}.$$

Notice that, if $N > 16 \log(pd^2)$, and $t^* = 4\pi C_M B \sqrt{\log(d^2 p)/N}$,

$$\frac{cN(t^*)^2}{4\pi C_M B^2} = 4c \log(pd^2) < 2c \sqrt{N \log(pd^2)} = \frac{cNt^*}{2\pi C_M B}.$$

Thus,

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{k=1}^n \frac{1}{T-1} X_{ki} (\mathbf{Y}_{\cdot m}^k)' \mathbf{E}_{\cdot \ell}^k \right| \geq 4\pi C_M B \sqrt{\frac{\log(d^2 p)}{N}} \right] \leq 6 \exp \left\{ -4c \log(pd^2) \right\}.$$

By applying the union bound, we obtain,

$$\begin{aligned} \mathbb{P} \left[\max_{\substack{1 \leq i \leq p; \\ 1 \leq m, \ell \leq d}} \left| \frac{1}{n} \sum_{k=1}^n X_{ki} (\mathbf{Y}_{\cdot m}^k)' \mathbf{E}_{\cdot \ell}^k \right| \geq 4\pi C_M B \sqrt{\frac{\log(d^2 p)}{N}} \right] &\leq 6 \exp \left\{ \log(pd^2) - 4c \log(pd^2) \right\} \\ &\leq 6 \exp \left\{ -3c \log(pd^2) \right\}. \end{aligned}$$

□

Proposition 5. For any $t > 0$, consider

$$\begin{aligned}\delta &:= \delta(t) = 7\sqrt{\frac{\eta B^2(\log(d^2p) + t)}{n}} + \frac{KB^2 \log(2n)(\log(d^2p) + t)}{n}, \\ \phi_1 &:= \inf_{\mathbf{x}} \lambda_{\min}(\Gamma_{Y|\mathbf{x}}(0)), \\ \alpha^* &= \phi_1 \lambda_{\min}(\Sigma_X)/2, \\ \tau^* &= 54\delta.\end{aligned}$$

Then, for all $t > 0$ such that $\alpha^* > 54s\delta$, we have that $\hat{\Gamma} = (\mathbf{I}_d \otimes \mathbf{W}'\mathbf{W})/N \in \text{RE}(\alpha^*, \tau^*)$ with probability at least $1 - 3e^{-t}$.

Proof. First, by Lemma 2, it suffices to show $\mathbf{W}'\mathbf{W}/N \in \text{RE}(\alpha, \tau)$. We denote $\Sigma_W = \mathbb{E}[\mathbf{W}'\mathbf{W}/N]$. To achieve this, the steps will be as follows: (i) show that $\mathbf{v}'\Sigma_W\mathbf{v} > K\|\mathbf{v}\|_2^2$ for some constant $K > 0$, (ii) use Lemma 5 to show that $|\mathbf{v}'(\mathbf{W}'\mathbf{W}/N - \Sigma_W)\mathbf{v}| = o_p(1)$, and (iii) apply Lemma 1 to derive the restricted eigenvalue property for $\mathbf{W}'\mathbf{W}/N$ with probability at least $1 - 3e^{-t}$.

First, we note that,

$$\frac{1}{N}\mathbf{W}'\mathbf{W} = \frac{1}{n} \sum_{k=1}^n \mathbf{X}_k \mathbf{X}_k' \otimes \left(\frac{1}{T-1} (\mathbf{Y}^k)' \mathbf{Y}^k \right)$$

Let $\mathbf{V} \in \mathbb{R}^{p \times d}$ be a matrix with columns $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_d]$. Let $\mathbf{v} = (\mathbf{v}'_1, \mathbf{v}'_2, \dots, \mathbf{v}'_d)' \in \mathbb{R}^{pd}$. Then, observe that,

$$\mathbf{W}^k \mathbf{v} = [\mathbf{X}_k' \otimes \mathbf{Y}^k] \mathbf{v} = \sum_{a=1}^p x_{ka} \mathbf{Y}^k \mathbf{v}_a = \mathbf{Y}^k \mathbf{V} \mathbf{X}_k$$

Using the same notation for $\mathbf{V} \in \mathbb{R}^{p \times d}$ and $\mathbf{v} \in \mathbb{R}^{pd}$, we have that

$$\begin{aligned}\mathbb{E} \left[\frac{1}{N} \mathbf{v}' \mathbf{W}' \mathbf{W} \mathbf{v} \right] &= \frac{1}{N} \sum_{k=1}^n \mathbb{E} [\mathbf{X}_k' \mathbf{V}' (\mathbf{Y}^k)' \mathbf{Y}^k \mathbf{V} \mathbf{X}_k] \\ &= \mathbb{E} \left\{ \mathbb{E} \left[\frac{1}{n} \sum_{k=1}^n \mathbf{X}_k' \mathbf{V}' \left(\frac{1}{T-1} (\mathbf{Y}^k)' \mathbf{Y}^k \right) \mathbf{V} \mathbf{X}_k \middle| \mathbf{X} \right] \right\} \\ &= \mathbb{E} [\mathbf{X}_1' \mathbf{V}' \Gamma_{Y|X_1}(0) \mathbf{V} \mathbf{X}_1] \\ &\geq \phi_1 \mathbb{E} [\mathbf{X}_1' \mathbf{V}' \mathbf{V} \mathbf{X}_1] \geq \phi_1 \lambda_{\min}(\Sigma_X) \|\mathbf{v}\|_2^2\end{aligned}$$

Now, we focus on providing concentration for $\sup_{\|\mathbf{v}\|_0 \leq 2s; \|\mathbf{v}\|=1} |\mathbf{v}'(\mathbf{W}'\mathbf{W}/N - \Sigma_W)\mathbf{v}|$. Note that,

$$\frac{1}{N}\mathbf{W}'\mathbf{W} = \frac{1}{n} \sum_{k=1}^n (\mathbf{X}_k \mathbf{X}_k') \otimes \left(\frac{1}{T-1} (\mathbf{Y}^k)' \mathbf{Y}^k \right).$$

Since the terms $\{(\mathbf{X}_k \mathbf{X}_k') \otimes [(\mathbf{Y}^k)' \mathbf{Y}^k]/(T-1)\}_{k=1, \dots, n}$ are independent, we may apply Lemma

5. To this aim, observe that for any $1 \leq i, j \leq d$ and $1 \leq \ell, m \leq p$, by Lemma 3,

$$\begin{aligned} \left| \left[\frac{1}{T-1} (\mathbf{X}_k \mathbf{X}_k') \otimes (\mathbf{Y}^k)' \mathbf{Y}^k \right]_{i\ell, jm} \right| &= \frac{1}{T-1} |X_{k\ell} X_{km} (\mathbf{Y}_{\cdot i}^k)' \mathbf{Y}_{\cdot j}^k| \\ &\leq \frac{B^2}{T-1} |(\mathbf{Y}_{\cdot i}^k)' \mathbf{Y}_{\cdot j}^k| \\ &\leq B^2 \Gamma_{Y|X_k}(0)_{ij} + O_p \left(\frac{C_M B^2}{T-1} \right) = O_p(B^2). \end{aligned}$$

On the other hand,

$$\begin{aligned} \max_{\substack{1 \leq i, j \leq d \\ 1 \leq \ell, m \leq p}} \frac{1}{n} \sum_{k=1}^n \mathbb{E} \left[\left(\frac{1}{T-1} X_{k\ell} X_{km} (\mathbf{Y}_{\cdot i}^k)' \mathbf{Y}_{\cdot j}^k \right)^2 \right] &= \max_{\substack{1 \leq i, j \leq d \\ 1 \leq \ell, m \leq p}} \mathbb{E} \left[\left(\frac{1}{T-1} X_{1\ell} X_{1m} (\mathbf{Y}_{\cdot i}^1)' \mathbf{Y}_{\cdot j}^1 \right)^2 \right] \\ &\leq \max_{\substack{1 \leq i, j \leq d \\ 1 \leq \ell, m \leq p}} B^2 \mathbb{E} \left[\left(\frac{1}{T-1} (\mathbf{Y}_{\cdot i}^1)' \mathbf{Y}_{\cdot j}^1 \right)^2 \right] \\ &\leq O_p(B^2) \end{aligned}$$

As a consequence, for appropriately chosen constants $K, \eta > 0$, by Lemma 5,

$$\left\| \frac{1}{T-1} \mathbf{W}' \mathbf{W} - \Sigma_W \right\|_{\max} \leq \delta := 7 \sqrt{\frac{\eta B^2 (\log(d^2 p) + t)}{n}} + \frac{K B^2 \log(2n) (\log(d^2 p) + t)}{n}$$

with probability at least $1 - 3e^{-t}$. From this, it can easily be shown that,

$$\sup_{\|\mathbf{v}\|_0 \leq 2s; \|\mathbf{v}\|_2 = 1} \left| \mathbf{v}' \left(\frac{\mathbf{W}' \mathbf{W}}{N} - \Sigma_W \right) \mathbf{v} \right| \leq 2s\delta.$$

Applying Lemma 1, it follows that, for any $\mathbf{v} \in \mathbb{R}^{pd}$

$$\left| \mathbf{v}' \left(\frac{\mathbf{W}' \mathbf{W}}{N} - \Sigma_W \right) \mathbf{v} \right| \leq 54s\delta \|\mathbf{v}\|_2^2 + 54\delta \|\mathbf{v}\|_1^2.$$

To conclude, observe that for any $\mathbf{v} \in \mathbb{R}^{pd}$

$$\begin{aligned} \left| \mathbf{v}' \left(\frac{1}{T-1} \mathbf{W}' \mathbf{W} \right) \mathbf{v} \right| &\geq |\mathbf{v}' \Sigma \mathbf{v}| - \left| \mathbf{v}' \left(\frac{\mathbf{W}' \mathbf{W}}{N} - \Sigma_W \right) \mathbf{v} \right| \\ &\geq (\phi_1 \lambda_{\min}(\Sigma_X) - 54s\delta) \|\mathbf{v}\|_2^2 - 54\delta \|\mathbf{v}\|_1^2. \end{aligned}$$

Thus, as long as $\phi_1 \lambda_{\min}(\Sigma_X)/2 > 54s\delta$, it follows that, $\mathbf{W}' \mathbf{W}/N \in RE(\alpha^*, \tau^*)$ for $\alpha^* = \phi_1 \lambda_{\min}(\Sigma_X)/2$ and $\tau = 54\delta$ and $\delta > 0$ is as defined above. $\square$

Theorem 2. Consider the MOD-VAR estimator $\hat{\Psi} \in \mathbb{R}^{d \times (dp)}$ defined by (35). Assume that Ψ^* is s -sparse, i.e. $\|\Psi^*\|_0 = s$, and we have equal sample size T for all n subjects. Let $N = n(T - 1)$. Then, there exist universal constants $K, \eta, c > 0$ such that the following statements hold. Let $t > 0$, and

$$\delta := \delta(t) = 7\sqrt{\frac{\eta B^2(\log(d^2 p) + t)}{n}} + \frac{K B^2 \log(2n)(\log(d^2 p) + t)}{n},$$

$$\phi_1 := \inf_{\mathbf{x}} \lambda_{\min}(\Gamma_{Y|\mathbf{x}}(0)),$$

$$\alpha^* := \lambda_{\min}(\Sigma_X) \phi_1 / 2,$$

$$\tau^* := 54s\delta,$$

$$C_M := \sup_{\mathbf{x} \in \mathbb{R}^p} \Lambda_{\max}(\Sigma_{\varepsilon}) \left(1 + \frac{1 + \mu_{\max}(\mathcal{A}_x)}{\mu_{\min}(\mathcal{A}_x)} \right),$$

$$Q(\beta^*, \Sigma_{\varepsilon}, B) := 4B\pi C_M.$$

Whenever the following inequalities are satisfied

$$\alpha^* > 32s\tau^* \tag{46}$$

$$N > 16 \log(pd^2) \tag{47}$$

$$C_M < \infty, \tag{48}$$

$$\lambda > 4Q(\beta^*, \Sigma_{\varepsilon}, B) \sqrt{\frac{2 \log(d) + \log(p)}{N}}. \tag{49}$$

our MOD-VAR estimator satisfies the consistency

$$\begin{aligned} \|\hat{\Psi} - \Psi^*\|_2 &\leq \frac{24\sqrt{s}Q(\beta^*, \Sigma_{\varepsilon}, B)}{\phi_1 \lambda_{\min}(\Sigma_X)} \sqrt{\frac{2 \log(d) + \log(p)}{N}}, \\ \|\hat{\Psi} - \Psi^*\|_1 &\leq \frac{144sQ(\beta^*, \Sigma_{\varepsilon}, B)}{\phi_1 \lambda_{\min}(\Sigma_X)} \sqrt{\frac{2 \log(d) + \log(p)}{N}}. \end{aligned}$$

with probability at least $1 - 3e^{-t} - 6e^{-3c \log(d^2 p)}$.

Proof. By (46), we have $\alpha^* > 32s\tau^* > 108s\delta$. By Proposition 5 we have $\hat{\Gamma} \in \text{RE}(\alpha^*, \tau^*)$ with probability at least $1 - 3e^{-t}$.

By (47), (48), and applying Proposition 4 we have that,

$$\|\gamma - \hat{\Gamma}\beta^*\|_{\infty} < Q(\beta^*, \Sigma_{\varepsilon}, B) \sqrt{\frac{2 \log(d) + \log(p)}{N}}$$

with probability at least $1 - 6 \exp\{-3c(2 \log(d) + \log(p))\}$. Therefore, Assumption 4 is satisfied.

Furthermore, note that by (49),

$$\lambda \geq 4Q(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B) \sqrt{\frac{2 \log(d) + \log(p)}{N}}.$$

Since the restricted eigenvalue condition is satisfied for $\alpha^* > 32s\tau^*$, and Assumption 4 holds, we have that:

$$\begin{aligned} \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 &\leq \frac{3\sqrt{s}\lambda}{\alpha} = \frac{24\sqrt{s}Q(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B)}{\phi_1 \lambda_{\min}(\Sigma_X)} \sqrt{\frac{2 \log(d) + \log(p)}{N}}, \\ \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1 &\leq \frac{12s\lambda}{\alpha} = \frac{144sQ(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B)}{\phi_1 \lambda_{\min}(\Sigma_X)} \sqrt{\frac{2 \log(d) + \log(p)}{N}}. \end{aligned}$$

□

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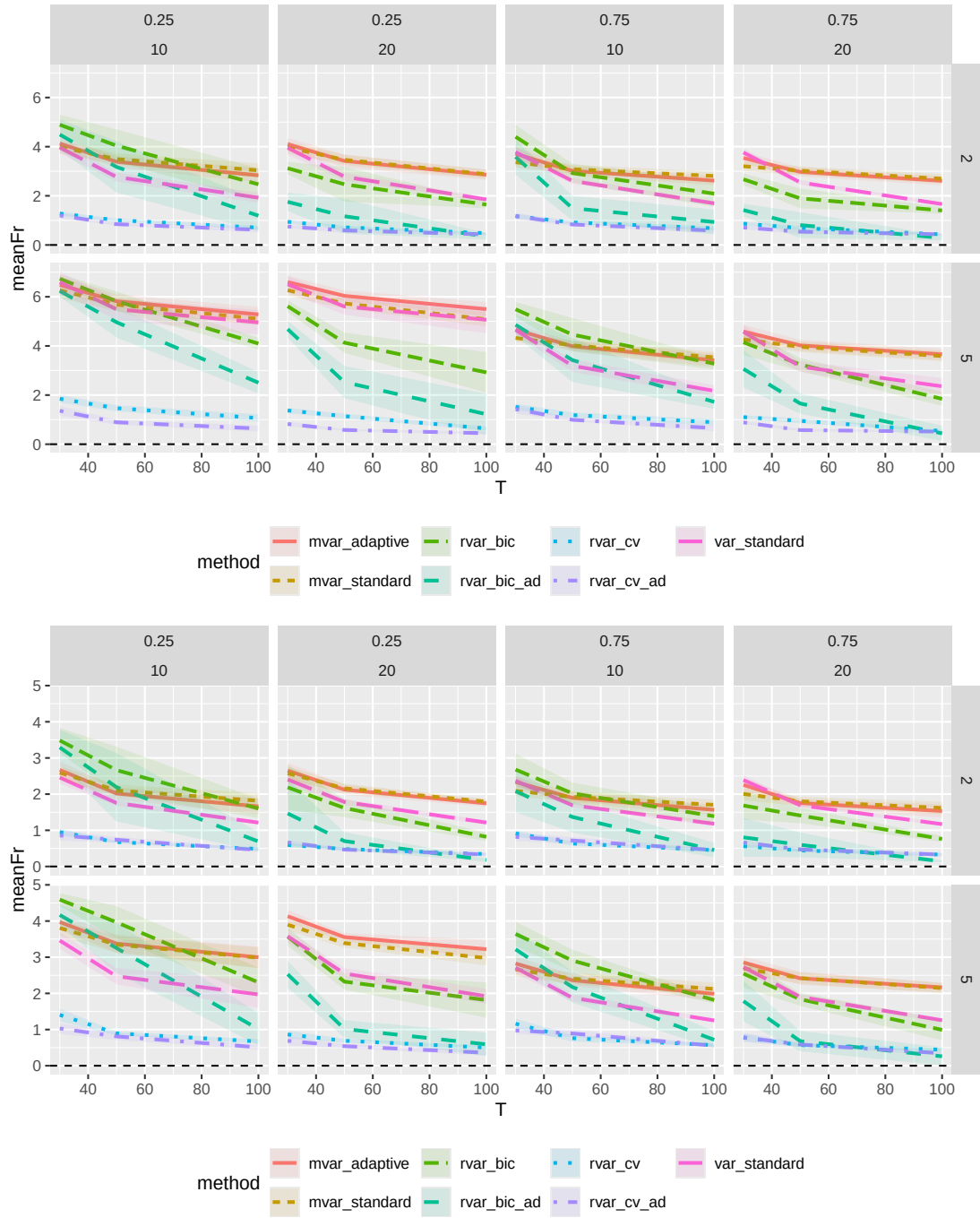


Figure 4: This is the caption for the inserted PDF page.